

A330

VER. 26.9.13

KneeBoard

 [Click for Update](#)

By *Flyingdeuk*

DOMESTIC

JAPAN

CHINA

SE ASIA

OCEANIA

MONGO, ME

Differences

FUEL

1 ENG TAXI

LP/HP Air

HP Air Start

Cold Temp

Cold Wx

ENG ON

ENG OFF

Deicing

Deicing

QRH

RESET

복행 GA 이후 / Go Around In flight

손님 여러분, 저는 기장입니다.

우리 비행기는 ① _____ 다시 상승했습니다.

- ① 기류가 불안정해
- ① 관제탑의 지시로
- ① 활주로가 육안으로 확인되지 않아
- ① 기술적인 이유로

[재접근 가능 시]

① _____, 잠시 후에 착륙할 예정입니다. 기다려 주셔서 고맙습니다.

- ① 기상 상태가 좋아져서
- ① 관제탑으로부터 다시 착륙을 허가 받아
- ① 관련된 문제가 해결되어

[Holding/Diversion 필요시]

이 점 양해해 주시기 바랍니다.

Ladies and gentlemen, this is the captain speaking.
We performed a go-around ① _____.

- ① due to unstable winds
- ① because we could not visually confirm the runway
- ① due to instructions from Air Traffic Control
- ① for technical reasons

[Re-approach]

① _____, and we expect to land shortly. Thank you for your understanding.

- ① The weather has improved
- ① We have now received clearance
- ① The situation has been resolved

[Holding/Diversion is required]

Thank you for your patience and understanding.



DOMESTIC

ICN

인천국제

GMP

김포국제

CJU

제주국제

PUS

김해국제

WELCOME PA (10분이상 지연 or 예상시 지연 방송)

손님 여러분, 안녕하십니까?

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저희 대한항공을 이용해 주셔서 대단히 감사합니다.

____ (국제)공항까지 비행시간은 ____시간 ____분 으로
예상됩니다.

비행 중에는 항공기가 갑자기 흔들릴 수도 있으니, 자리
에 앉아 계실 때에는 항상 좌석벨트를 매주시기 바랍니
다. (저희 승무원 모두는)

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고맙습니다.

Good morning (afternoon /evening), ladies
and gentlemen.

This is captain last name speaking.

Welcome aboard Korean Air.

This flight is bound for ____ (international)
airport and our flight time is ____ hours(s) and
minutes.

For your safety, keep your seatbelts fastened
while you are seated.

Thank you for choosing Koreanair.

Please enjoy your flight.



도착방송

도착 방송 (5시간이상, 40분전)

출발지 기준 2200-0800 Quiet Hour

동남아 지역 난기류 대비 기내 서비스 절차

- 6시간 미만 중거리만 1시간 전 음료서비스

- 40분전 서비스 회수

- APP PA이후 CREW 착석 강조

손님 여러분, 저는 기장입니다.

우리 비행기는 앞으로 약 (40)분 후에 ____국제공항에 착륙 예정입니다.

현재 공항의 날씨는 ㉠____, 기온은 섭씨 ____도 입니다.

㉠ 맑으며

㉡ 눈이 오고 있으며

㉢ (다소)흐리며

㉣ 안개가 끼어 있으며

㉤ (이슬)비가 내리며/소나기가 내리며

㉥ 황사가 있으며

㉦ 바람이 불고 있으며

지금 이곳의 시각은 ____월 ____일 ____요일, 오전(오후) ____시 ____분
입니다. 고맙습니다.

Ladies and gentlemen, this is the captain speaking.

We expect to land at ____ international airport in about
(40) minutes.

The current temperature at ____ is ____ degrees Celsius,
or ____ degrees Fahrenheit (NAVBLUE 참조)
and it is ㉠____.

㉠ (mostly) clear

㉡ snowing

㉢ (partly) cloudy

㉣ foggy

㉤ drizzling / raining

㉥ hazy or smoggy

㉦ windy

The current time is ____ : ____ a.m(p.m), on (day-of-the-
week), (month)(date).



JAPAN

HND

도쿄 하네다

KIX

오사카 간사이

NRT

도쿄 나리타

FUK

후쿠오카

NGO

나고야 주부
(Chubu Centrair)

CTS

삿포로
신(New) 치토세

KOJ

가고시마

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Please enjoy your flight.



CHINA

HKG

홍콩

PEK

베이징 소우뚜

WUH

우한 텐허

CAN

광저우 바이윈

XIY

시안 시옌양

TSN

톈진 빈하이

TAO

칭다오 자오둥

SZX

선전 바오안

PVG

상하이 푸둥

SHE

선양 타오셴

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SE ASIA

BKK

방콕 수완나폼

SIN

싱가포르 창이

DAD

다낭

SGN

호찌민 탄소넛

CNX

치앙마이

DPS

덴파사르 발리 응우라라오

KTI

프놈펜 테초

TPE

타이페이 타오유엔

RGN

양곤

KUL

쿠알라룸푸트

CXR

냐짱(나트랑)/깜라인(깜란)

PQC

푸꾸옥

HAN

하노이 노이바이

MNL

마닐라 니노이아키노

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도착방송

OCEANIA

SYD

BNE

시드니
킹스폴드 스미스

브리즈번

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도착방송

MONGO, ME ASIA

UBN

DXB

KTM

울란바타르 칭기스칸

두바이

카트만두 트리부반

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도착방송

RKSI(ICN) 23ft

KE ICN 131.5
(TOBT 조절)

UTC+9

DCL -10min
TOBT 5분차이시 CTC COMM

ARR
TL140

Tx PT GUKDO, OLMEN, REBIT, KARBU
FL180,170,160 예상

33L/R(334)	OLMEN	xE	ENPIL	ILS
34L/R(334)	GUKDO	xY	PAMBI	ILS
15L/R(154)	REBIT	xC	BITIM	ILS
16L/R(154)	KARBU	xH	MUNAN	ILS Z

NCN 113.8 33L INLL 109.3 33R INRR 108.9 15L ISLL 111.9 15R ISRR 109.1
WNG 112.9 34L IRFN 109.95 34R IRKN 108.1 16L IRKS 110.35 16R IRFS 108.35

SPD

8NM 180kts, 5NM 160kts

FIX

YJU R271, P73 /2, P518 R068, R278

33R : C5(5501'), C6(6683'), 33L : B4(7463'), B5(8513')
15L : C4(5495'), C5(6227'), 15R : B3(7454'), B2(8641') **HIRO**
34L : P7(5600'), P8(6578'), 34R : N4(6876'), N5(8507')
16R : P6(5597'), P5(6574'), 16L : N3(7043'), N2(8444') **HIRO**

Parallel TWY 10KTS이상, R17 Max 15KTS

DEP

NADP

TA140

RWY 33/34 NADP 1, RWY 15/16 NADP 2

33L/R(334)	BOPTA	xE/A	ALT ATC(5500)
34L/R(334)	EGOBA	xY	ALT ATC
15L/R(154)	NOPIK	xC	ALT 5000
16L/R(154)	OSPOT	xH	ALT 5000

EO

33L/R : NC05L/R R242,
34L/R : EO34L/R R242

Parallel TWY 10KTS이상, R17 Max 15KTS



PA

RKSS(GMP) 59ft

KE GMP 131.15

UTC+9

DCL -15min

TOBT 5분차이시 CTC COMM

ARR
TL140



RWY 32L Landing

(06:00-09:00L/12:00-15:00L/18:00-21:00)

32L/R
(324)

OLMEN
GUKDO
KARBU
REBIT

xT
(xQ)

BUMSI

ILS

14R(144)

xU
(xZ)

DOKDO

ILS

NADP

32 : ILS 7 DME L/G D/N, FAF L/D FLAP

14 : LOC CAPT L/G D/N, FAF L/D FLAP

NAV

KIP 113.6 32L IKMO 108.3 32R ISKP 110.7 14L
ISEL 109.9 14R IOFR 108.7

FIX

YJU R271, P73 /2, P518 R068, R278

32L : D3(6532'), E2(9117'), 32R(No CL L/T) : E1(6614')

14R : C1(6578')

TWR -> GND -> APRON all by ATC

RTE 1 : D3-P6-N1 H.S P5 (BOKUM, NUNPI, SAMKU)

DEP
TA140



RWY 32R Takeoff

(06:00-09:00L/12:00-15:00L/18:00-21:00)

Walk Around

비밀번호 없어짐, 정비사가 열어줌

NADP

FX-10도, CLB, CONF 1+F, ACC 4000FT

32L/R
(324)

BULTI
EGOBA
NOPIK
OSPOT

xT
(xQ)

ALT 5000

14R(144)

xU
(xZ)

ALT 6000

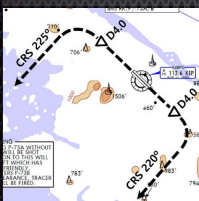
EO

32L/R : EO32L/R, R225

14L/R : EO14L/R, R220

TWR -> GND -> APRON all by ATC

(BOKUM, NUNPI, SAMKU)



PA

RKPC(CJU) 119ft

KE CJU 129.4

UTC+9

DCL -10min

PREFLT

외부점검 비밀번호 정비사 문의

ARR **AFT Merge PT 220kts, DCT IAF 195kts, FAF**
TL140 160kts

07**(066)** **DOTOL xP** YUMIN ILS Z

25**(246)** **DOTOL xM** DUKAL ILS Z

RNP Y 07/25 (No LPV), RNP Z 07/25 AR (RNP 0.11)

NAV **YDM 109.0 07 ICJU 109.9 25 ICHE 111.3**

FIX YJU R271, P73 /2, P518 R068, R278

07 : **P6(5176'), P5(5882')-HIRO**, P4(6840'-Req ATC)

25 : **P7(5219'), P8(5882')-HIRO**, P10(7524'-Req ATC)

HIRO in Force CHK, HST 40KTS

Entering Rapid TWY CTC GND 121.675 (STOP X)

DEP **07 : Passing G4 CTC TWR**

TA140 25 : 31 Holding PSN on P,E1,2,3 CTC TWR

Walk Around 비밀 번호

07**(066)** **KAMIT xE** ALT 10000

25**(246)** **KAMIT xW** ALT 10000

EO 25 : YDM 246/3, R290

Caution RW31 Holding Sign & STOP BAR L/T

HIRO in Force Prepare Intersection DEP



PA

RKPK(PUS) 13ft

KE CJU 129.4

UTC+9

DCL -5min

ARR **Tail Wind 10kts, 36R 360000lbs 제한**
TL140 **Circling Procedure Next Page**

36L/R **KEVOX x** MASTA ILS
(002) 9DME L/G D/N 8DME L/D FLAP

18L/R **GAYHA x** MSATA VOR Circling
(182)

NAV **KMH 113.8 36L IKMA 108.5 36R IKHE 109.5**

36L : **C4 (6299')**, C2(7795') 36R : E3(5866'), E2(7339')
18R(**DIS**):C6(5770'),C7(6824') 18L:E4(5882'),E5(8792')

Vacate **C3, C4** by ATC Only. Max Taxi SPD 20kts
C2 HOLD SHORT 가까움 (VACATE TAXI SPD)

DEP **Modified NADP**
TA140 **CLB 1500, 360000lbs TOGA, CONF 1+F**

Walk Around 비밀번호

36L/R **SOORO x KALOD Tx** ALT ATC
(002)

18L/R **GIMHAE x** ALT 5000
(182)

EO 36 : KMH R091, R271, R185 R290

RWY36 400ft Manual L/H TURN
Max Taxi SPD 20kts



Circling
Next Page



PA

A330 PUS VOR C 18L/R

ARRIVAL to RKPK ↔			SEC ARRIVAL to RKPK ↔		
APPR	VIA	STAR	APPR	VIA	STAR
VOR18R	GAYHA	GAYHA3	18R	NONE	V18R
APPR		TRANS			TRANS
<VIAS		KALOD			NONE

Input PERF -> SEC COPY

CLR CP18L/R, DISCONT
MDA :[], L/D CONF chk

FIX : KMH 280(Base Turn), 284(Missed App)



Missed App

Before MAP : L/H Turn **KMH 284** OUTBD to GAYHA **GA 6000ft**

After MAP : Before Abeam RWY HD310 6000ft -> RDR Vector

After Abeam RWY R/H Turn RWY -> HD180, 6000ft -> RDR
Vector

A330 PUS LOC 36L/R Circling 18L/R

F-PLN			SEC F-PLN	
APPR	VIA	STAR	APPR	STAR
LOC36L	KEVOX	KEVOX x	18L/R	V18L/R
		TRANS	(MDA: Blank)	
		MASTA	(LDG CONF chk)	

CI36L(CF36R) 3500 FI36L(FF36R) 2100

FIX : KMH 280(Base Turn), 310(Missed App)



Missed App

Before MAP : L/H Turn **KMH 284** OUTBD to GAYHA **GA 6000ft**

After MAP : **Before Abeam RWY** HD310 6000ft -> RDR Vector

After Abeam RWY R/H Turn RWY -> HD180, 6000ft -> RDR Vector

RJTT(HND) 21ft Terminal 3

None

UTC+9

DCL -15min

PREFLT

북풍시 34L 주로 사용, 남풍시 22/23사용

**ARR
TL140**

SPENS 220

STAR XAC Night-APP ILS xx Y 1400~

ILS xxZ : 180kts, 160kts limit APP Chart

22/23 LDA, VOR 16L/R Tailored Procedure

34L/R
(338)

XAC xK/H
XAC xA

KAIHO/CACAO
ARLON

ILS X, VIS
ILS Z

22(223)

XAC 1B

BACON

LDA W (RNV22-W)

16L/R
(158)

XAC R

NATTY/SANDY

RNP (RNV16R/L)

23(231)

XAC 2B

DANON

LDA W (RNV23-W)

NAV

TTE117.4 34L IHA 111.7 16R ITA 111.55 34R

ITC 108.9 16L IOC 111.95

22 IAD 108.1 23ITD 110.5 (LDA 22 IK, 23 ITL)

34L: **L12(6515), L13(7165)**, 16R(DIS): L5(5147), L3(6361)

22 : B4(6207), B3(6830), 23: D5(5072), D3(6391)

34R(DIS): C10(5780), C11(7270), 16L(DIS): C6(5925), C4(7004)

Caution L12,13 then TWY, Gate 105P Procedure

**DEP
TA140**

xx B/C 2200-0230Z 0600-1000Z

34R BEKLA KAIJI 230, TO FLAP SPD Until TORAM

05 OPPAR : Reduce SPD ARC TURN

PA

ALL

BEKLA x
OPPAR x

ALT ATC

EO

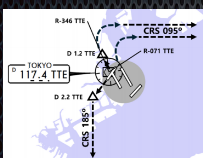
34L : TTE 346/1.2, R095, 34R : TTE R071, R095,

22 : TTE 223/2.2, R185

RWY05 RTE 5 TAXI Chart

- DEP **Freq ATIS**

PA



LDA
Next Page



A330 HND LDA 22 W

F-PLN

APPR

STAR

TRANS

FIX

RNV22-W

XAC 1B

BACON

TTE R015

CHK IKL, IKL /8 180kts, /3 160kts, TTE 015 (GA)



고려사항

- Procedure SPD 지켜야 Traffic 간격 유지됨. 무게에 따라 -4도 증속가능성 있음.
- S/B 사용하거나 강하각을 조절하는 것이 필요함.

Missed App

R/H Turn **HME 015** OUTBD (PULL TRK)
(DCT KASGA Radial In 196, MAN NAV)

A330 HND LDA 23 W

F-PLN

APPR

STAR

TRANS

FIX

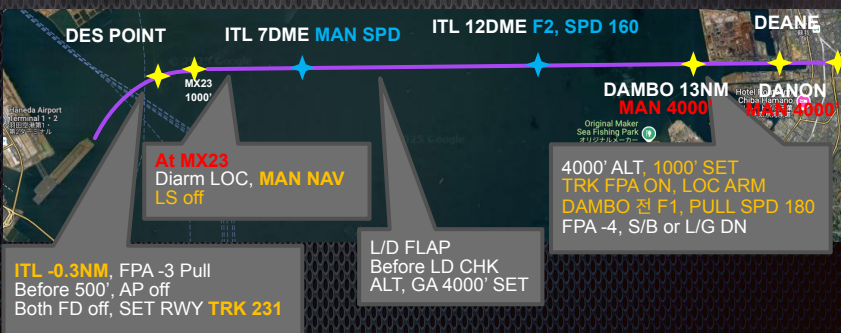
RNV23-W

XAC 2B

BACON

TTE 178

CHK ITL, ITL /12 180kts, /7 160kts, TTE 178 (GA)



Missed App

L/H Turn **HME 177** OUTBD (PULL TRK)
(DCT UTIBO Radial In 357, MAN NAV)

RJBB(KIX) 17ft Gate 9

KE KIX 130.95

UTC+9

DCL -15min

ARR
TL140

SAEKI 170, RANDY 150

06R/L
(059)

ATMUG B
ATMUG A

BERRY
ATVID

ILS Z

After 2500ft L/G DN, After 1500ft L/D FLAP

24L/R
(239)

ATMUG M/C
ATMUG D/E

MAYAH
D/E Night

ILS Z

NAV

KIE 111.6 06L IKJ 108.7 06R IKD 108.1
24L IKN 110.7 24R IKW 108.5

06R : A7(5137'), A6(6938'), 24L : A8(5269'), A9(6976')

06L : B8(5160'), B6(6751'), 24R : B7(5318'), B9(6751')

24L Downslope

TAXI RTE 1(via J4), 2(via J3)

PA

DEP
TA140

SOUJA Tx

Walk Around

안에서는 정비사가 밖에서는 비밀번호

06R/L
(059)

ALT ATC(9000)

RINKU x - SOUJA Tx

24L/R
(239)

ALT ATC(9000)

TAXI RTE 1(via J4), 2(via J3)

- DEP 119.2

PA

IN
OUT

주황색 Landing Permit



RJAA(NRT) 135ft Gate 21-27

KE TOKYO 131.7

UTC+9

DCL -15min

PREFLT

L/D은 북쪽 활주로, T/O는 남쪽 긴 활주로
TAXI GND Marking 잘 봐야함

ARR
TL140

HAKKA 330, YAGAN 240, LIVET 210,
SWAMP 150

34R/L
(338)

SWAMP E
SWAMP T

ELGAR
TYLER

ILS Z

L/G DN before ILS DME 14/12NM, 4NM L/D FLAP

16L/R
(158)

SWAMP G
SWAMP N

GEMIN
NORMA

ILS Z

ILS DME 4NM L/D FLAP

NAV

NRE 117.9 16L ITM 110.7 16R IKF 111.5
34L IYQ 111.9 34R ITJ 110.9

16L : B6(6433'), B7(7017'), 34R : B4(5849'), B2(6778')
16R : A6(6076'), A7(7624'), 34L : A5(6167'), A4(7641')

Arrival TAXI RTE in JEPP (No Numbering)

PA

DEP
TA140

ENPAR Tx

Walk Around 기계식 버튼 비밀 번호

16R/L
(158)

TETRA x - ENPAR Tx

ALT ATC

34L/R
(338)

Check ALT

ALT 7000(ATC)

Departure TAXI RTE 1, 2, 3, 4

- DEP 124.2

IN
OUT

주황색 Landing Permit

PA



RJFF(FUK) 30ft Gate 53-59

KE FUK 132.05

UTC+9

DCL -15min, Voice -5min

PREFLT

배풍 10kts까지 16사용, ILS 는 저시정에서만 사용

**ARR
TL140**

PAVGA 13000

Hold W of IKE publish (Confirm FMS DATA)

**16L/R
(158)**

ENTIX

**RNP
LOC**

**RDR Vector
Around IKE**

**34R/L
(388)**

**R34RV
KIGUS**

VIS (**RWY34**)
ILS, RNP

NAV

DGC 114.5 (no A/P) 16L IFO 111.7

16R IFN 110.95 34R IFF 108.9 34L IFW 110.35

16L : C6(5475'), C7(6420'), 34R : C4(5124'), C3(6348')

16R : W7(5577'), W8(6696'), 34L : W4(5764'), W3(6466')

VIS 34 (Tailord Proc) : After IKE – RDR Vector Downwind – 1800ft – RWY Insight 1500ft – Before L/D CHK Complete before base (Do not Extend Downwind due Terrain)

**DEP
TA140**

Consider W2, W9 Intersection DEP

PA

Walk Around

**16R/L
(158)**

ALT ATC(10000)

HAKATA xx

**34L/R
(338)**

L/H Turn RDR Vector

ALT ATC(10000)

EO

16L/R : DGC 158/20 R240

Caution 34L APCH HOLD LINE, RWSL in Operation

Initial CTC TWR, “Ready for departure”

- DEP 127.9



PA

RJGG(NGO) 12ft Gate 19-24

SWISSPORT OPER 132.05

UTC+9

DCL -15min

ARR
TL140

SAMON 290, MARIA 130

36
(356)

CHESS(CARDS)
SOUTH

PROBE

ILS Z

After 1500ft L/D FLAP

18
(176)

CHESS(CARD)
NORTH

QUEST

ILS Z

After 3000ft L/G DN & L/D FLAP

NAV

CBE 117.8 18 ICY 109.7 36 ICX 111.9

36 : A6(5213'), A7(6525'), A8(7837')

18 : A5(5393'), A4(6528'), A3(7841')

Caution STOP LINE, Yellow Ramp Line, VDGS!!

PA

DEP
TA140

TANGO Tx

Walk Around 기계식 버튼 비밀 번호

36
(356)

ALT ATC(7000)

OUMI x - TANGO Tx

18
(176)

ALT ATC(7000)

APU Start 30min, Prepare Intersection T/O

- DEP 120.0

PA

IN
OUT

저녁식사후 하기 - 주황색 Landing Permit



RJCC(CTS) 70ft Gate 63-71

COM None

UTC+9

NO DCL, Voice -5min

PREFLT

강설 주의, TAXI 불가시 Req Towing

ARR
TL140 01R : IDEMI 150
19L : NAVER 170

01R/L
(003) **NAVER xxY** **YOTEI** ILS Y **3000**
MUKAWA xxB Not YOSEI ILS Z **2000**

19L/R
(183) **NAVER xxP** **PUPIG** ILS Z
NAVER xxK KAORY **CAT III**
19R only

NAV **CHE 116.9** **01R ICH 110.75** **19L ICM 109.35**
01L ICN 110.9 **19R ICS 111.5**

01R : B4(5278'), B3(7047'), 19L : B8(5177'), B9(7119')
01L : A5(5538'), A4(6961'), 19R : A7(5390'), A8(6873')

Do not Cross 01L/19R After L/D (No TWY)
TAXI to Gate Via D(J) or G

PA

DEP **Deicing at the Gate**
TA140 **APU 30분전**

Walk Around 기계식 버튼 비밀 번호

01L/R
(003) **DALBI xx** ALT ATC
SAVIT xx

19R/L
(183) **SOSHU xx** ALT ATC

R/H Turn DCT HWE - Confirm R/H Turn ND

- DEP 124.7

PA



RJFK(KOJ) 891ft

Gate 2,3,4

Japan Air KOJ 131.0

UTC+9

DCL -5min (0700-2200L)
Monitor 121.8

PREFLT

Caution Volcanic Ash Information Before DEP

ARR
TL140

IKE FL260, KAZSA FL160 **TERR RW16 Final**
RW16 : Keep Manage NAV Until RWY Insight

34(337)

SIMAZ NORTH

SIMAZ

ILS Z

16(157)

OIDON

OIDON

RNP

NAV

KGE 115.7 34 IKG 111.7

34 : T5(6387'), T6(7637', 90도), 16 : T3(5990', 90도), T2(7988', 90도)
RW16 DownSlope

White Solid Line = Ramp Area
Follow Marshaller

PA

DEP
TA140

Req Planned ALT in DCL : P340 in RMK

Walk Around

34(337)

ATRUK x ASHIKITA Tx

ALT ATC(9000)

16(157)

Maintain RWY H/D

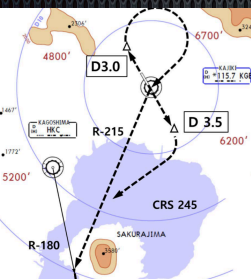
ALT ATC(9000)

EO

34 : KGE 337/3 R215, 16 : KGE 157/3.5 R215

- DEP 119.4

PA



ZGGG(CAN) 50ft Bay 147

China Southern 132.0

UTC+8

DCL -10-30min
NO Read Back

PREFLT

여권 번호만 확인 제출X

**ARR
TL108**

CAN CTL, APP Freq no Chart, Low GA ALT
Short PTN AFT IAF, RW20 Windshear

**02R01L
(017)**

GG611/GG612 ILS Z

OLPAB xx RNAV

**20L19R
(197)**

GG511/GG513 ILS Z

SPD

8NM 180kts, 6NM 160kts

NAV

02R IDM 108.5 01L IGW 109.5 20L IXL 111.9
19R IKU 108.95 02L IBB 110.35 01R IOO 109.3
20R IAA 110.75 19L IPP 111.5 POU 114.1(no A/P)

02R:Y10(6227'),Y8(7263'), 20L:Y12(6217'),Y14(7240')

01L:N5(6453'),N4(7755'), 19R:N8(6423'),N9(7742')

Caution Cross RWY Intermediate TWY

STD Taxi RTE in JEPP(GREEN x), EAT 3/4 in JEPP

APU off Procedure, FLAP 0 for Refuel

PA

DEP

TA89

Consider Extra Fuel Delay/CRZ LVL

Walk Around

로톤다 엘베이용 비번 입력 V 누르고 엘베버튼 누름

**02L01R
(017)**

ALT ATC(1200m)

BOVMA xx RNAV

**20R19L
(197)**

ALT ATC(1200m)

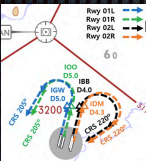
EO

01L/R IGW/IOO D5.0 R205, 02L/R IBB/IDM D4.0/4.3 R220

STD TXY RTE in JEPP(Yellow X) Report RWY on DEP

- DEP 119.7(DEP RWY)

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
			
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZGSZ(SZX) 13ft Bay 301-319

SZX Dispatch 131.3

UTC+8

DCL AVAILABLE

PREFLT 여권만 제출

ARR **SPD Restriction JEPP**
TL108 **Caution Freq CHG & RDR Vector (TERR!!)**

1516R **SAR 9/8 XA** SZ462 RNP ILS Z
(156)

3334L **SAR 9/8 ZA** NLG RNP ILS Z
(336)

SPD 30NM 220kts, 8NM 180kts, 5NM 160kts

NAV 15 IQJ 111.3 16 ISZ 108.1 33 IMH 110.7
34 IBA 109.1 (no A/P VOR)

16R : H6(5449'), H8(6676'), 34L : H6(5413'), H5(6561')
15: C8(6725'), C9(8152'), 33 : C5(6548'), C4(8175')

Be carefull RWY 16L then TWY Q,R

CTC Apron Before Entering, MAX wingspan in JEPP
APU OFF but Remain

PA

DEP
TA89 **Consider Extra Fuel Delay/CRZ LVL**

Walk Around 로툰다 문 앞에서 카메라 누르고 ID + 얼굴 보여주면 열림

1516L **IDU 9WD/XD** ALT ATC(1200m)
(156)

3334R **IDU 9YD/ZD** ALT ATC(1200m)
(336)

EO 15 : IQJ 155/3.5, R205, 16 : ISZ 155/3.5, R205



DEP ATIS (AUTO) Report DEP RWY

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZYTX(SHE) 198ft Bay 137-139

China Southern Dispatch
131.5

UTC+8

DCL -40min, Read Back
Voice -10min

PREFLT 여권 제출, GD Sign

ARR
TL118

CLR Limit TOSID Late Handoff to SHE
Around TOSID - Present TRK,H/D - CTC
SHE

06(056)

TOSID 62A, 61A

TX504

ILS Z

24(236)

TOSID 72A, 71A

TX603

ILS Z

NAV

SEY 114.1 06 IPU 110.5 24 IKY 110.3

06 : D(6210'), C(7854'), 24 : J(6227'), K(7864') Prefer Exit

06 : HP09(08), 24 : HP04 Follow Me Car

Normally Remain Parking Brake SET!!

APU off Procedure (GND Air Cord & GPU)

PA

DEP

TA98

A2, A8 Intersection T/O not Avail A330

ADT = CTOT See Enroute ATIS

Walk Around L1 비번없이 나가면 됨

06(056)

TOSID 61,62D

ALT ATC(900m)

24(236)

TOSID 71,72D

ALT ATC(900m)

"P/B follow GND Maintenance"

Be Careful "Hold Short CAT I Hold Line" Find Sign!!

Maintain Present TRK/HDG Join A588 (CRS 217)

Or Offset 3NM R/H of TRK

CTC APP without TWR Instruction

APP 121.225

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZBAA(PEK) 116ft Bay 205-208

Air China Beijing 132.0

UTC+8

DCL -40min, Voice -20min
COBT/STD 15분 차이 CTC COM

PREFLT

여권 번호만 확인

ARR
TL118

RWY 01/19 Main L/D (RWY36L/18R)

0136L
(001)

DUMAP xZ

AA141

ILS Z 01
(Y 36L)

1924
(181)

DUMAP xG

AA261

ILS Z 19
(Y 18R)

SPD

TMA Max 280, 8NM 180, 6NM 160

NAV

PEK 114.7 01 INJ 108.5 19 ISZ 108.9 36L IDK 111.7
18R ILG 110.3 36R IQU 111.55 18L IOR 109.3

01 : Q5(5223'), Q6(7024'), 19 **3.2** : Q4(5298'), Q3(7103')
36L : P6(6276'), P7(7719'), 18R : P3(6223'), P2(7552')

APU off Procedure (GND Air Cond' & GPU)
Standard TAXI RTE in Jeppesen Chart

PA

DEP
TA98

RWY 36R/18L Main T/O(Intersection W2, W7)
COBT on ATIS "Enroute", Bad Wx DOTRA SID

Walk Around

36R01
(001)

MUGLO xX/Y
Maintain RWY TRK

ALT 2100m(1800m)

18L19
(181)

MUGLO xG/J

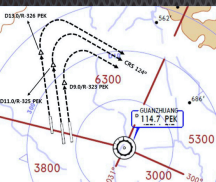
ALT ATC(1500m)

EO

36R : PEK 325/11, 36L : PEK 326/13
01 : PEK 323/9 -> R124

- APP 124.4

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZHHH(WUH) 115ft Bay 331-335

COM None

UTC+8

**DCL -30-10min
READBACK!!**

PREFLT 여권 번호 확인 및 제출, GD Sign

ARR **WUH CTL Rep HLxxxx**
TL118 **RDR CTC 60NM Prepare STAR CHG**

0405R **PONUD 01T** HH310
(046) **PONUD 05T** HH357 ILS Z

1924 **PONUD 02T** AVLEN ILS Z
(246)

SPD Intercept Final to 8NM 180kts

NAV **WHA 112.2 04 IHN 109.3 22 ITS 108.5 05R IMU 111.1**
IQD 111.35 05L IWF 111.5 23R IUT 111.5
04 : B8(5557'), B9(6578'), 22 : B5(6738'), B4(7939')
05R : F10(5475'), F11(6597'), 23L : F7(5341'), F6(6545')

Follow Me Car TWY K/J
APU off Procedure (GND Air Cond' & GPU) **PA**

DEP **Readback RWY, ALT to CLR**
TA98 **Report Ready for P/B to CLR**

Walk Around 2층에서 엘베타고 내려가서 비번입력

0405L **BIVIP 7S** ALT 4800m/ATC(900m)
(046)

2223R **BIVIP 6S** ALT 4800mATC(900m)
(246)

Uncommanded Turn RNP DEP

- APP 119.575

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
			
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZLXY(XIY) 1572ft Bay 556-561

Airport Oper Center 132.0

UTC+8

DCL -30-10min
READBACK!!

PREFLT 여권 번호 확인 및 제출, Custom 작성, GD Sign

ARR **SPD REST in REF Page**
TL118 **Req ILS APP instead of VIS APP**

05LR **LOVRA xZ/Y** XY906 RNAV ILS Z
(052)

23RL **LOVRA xW/Y** XY801 RNAV ILS Z
(232)

SPD ZS/FNH 220kts, XY914/918/807/810 200kts,
XY958/919/859/805 180kts, 5NM 160kts

NAV **LCZ 109.0 05L IQU 108.55 23R IRA 111.7,**
05R IXW 109.3 23L IAQ 111.1

05L : A3(6778'), A2(9032'), 23R : A6(5544'), A7(6512')

05R : D4(5613'), D3(7322'), 23L : D5(5646'), D6(7408')

Follow Me Car, CTC Apron Before Gate in Taxi RTE in JEPP
"Closing to xx TWY, apply to CHG to xx Freq"

PA

DEP **Apply NOTAM TO Performance**
TA98 **ADT = CTOT**

Walk Around

05LR **WJC xW** ALT ATC 1500m
(052)

23RL **WJC xX** ALT ATC 1500m
(232)

PushBack GREEN/BLU used

- APP 119.9

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

VHHH(HKG) 28ft Bay N5-N12

HAS FLT Dispatch 131.6

UTC+8

DCL -20min
5분 차이시 CTC Comm

PREFLT

Deicing 유사 Delay시 시간 초과 가능성 있음

ARR
TL110

ENPET 260 (1500-1659Z RWY 07R/25L X)
Terminal Tx RTE + STAR Chart
RWY 25R After TOPUN - APP Mode

07LRC
(074)

ABBEY xA
SIERA xA/C

LIMES

ILS

25RLC
(254)

ABBEY xB
SIERA xB/D

RIVMI

RNAV Tx ILS 25R
ILS 25L

SPD

Confirm Chart SPD vs ATC SPD (ATC First)

NAV

SMT, 07L IZSL, 07R IZSR, 07C IZSC, 25R ITFR,
25L ITFL, 25C ITFC

07L : C7(5882'), C8(7194'), 25R : C6(5882'), C5(7211')
07R : J7(6916'), J8(7998'), 25L : J5(6916'), J4(8192')
07C : A7(5692'), A8(7208'), 25C : A6(5127'), A56673')

Gate In Solid line For Wide Body, APU OFF Proc

DEP

SID + Terminal Tx RTE Chart

TA90

SID - Tx RTE Chart Many SPD Restriction

PA

Walk Around

NADP 2

THR RED : 1000ft, ACCEL ALT : 1000ft

CLB SPD : GREEN DOT PRESEL

1000' NO LVR CLB - FLAP 0 - THR CLB - MAN SPD

07CRL
(074)

DALOL xE/C/A
(Z/Y/X RF LEG)

ALT 5000 **NADP 1/2**

25CLR
(254)

DALOL xB/D/F

ALT 5000 **NADP 2**

EO

07L: IZSL074/3,R095, IZSL/9.1, 25R: ITFR254/10,R156
07C: IZSC074/2.7,R080, IZSC/6.2, 25C: ITFC254/10,R156
07R: IZSR074/6.2,R080, MAX200kt, 25L: ITFL254/6,R145

- DEP 123.8

PA



ZBTJ(TSN) 6ft Bay 105-107

Air China TSN 132.0

UTC+8

DCL -30min, Voice -10min
READBACK!!

PREFLT

ARR
TL118

Missed APP 1970ft Initially

16LR
(161)

DUMAP xK/R/H

TJ960
TJ919

ILS

34RL
(341)

DUMAP xQ

TJ841

ILS

SPD

6NM 160kts (JEPP REF)

NAV

TAJ 112.1 16R IJS 110.9 34L ICG 110.5
16L ICU 109.7 34R IKD 111.5

16L :W3(6269'),W2(9809'), 34R :W7(6443'),W8(7591')
16R :B4(5177'),B3(7191'), 34L :B5(5183'),B6(7201')

Follow Me Car on D, TAXI SPD Max 27kts

PA

DEP
TA98

Caution 600m Level Off - SPD Increase

Walk Around

16RL
(161)

MUGLO xT

ALT ATC(600m,2000')

34LR
(341)

MUGLO xW/R

ALT ATC(600m,2000')

16R : Do not pass A11

Confirm Parking Brake Release Before Push Back

- APP 119.27

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
			
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

ZSQD(TAO) 30ft Bay 131-141

Comm None

UTC+8

**DCL AVAIL, Voice -10min
READBACK!!**

PREFLT 여권 번호 확인 및 제출, Custom Form

ARR **AFT AGAVO - IKEKA - LATUX**
TL118 (WEH 훈련시 RE-RTE됨)

3534
(350) **LAT 91A/01A** **QD700** ILS Z

1716
(170) **LAT 81A/11A** **QD305** ILS Z

SPD 6NM 160kts

NAV **JDG 114.45 35 IQX 109.75 34 IIN 108.55**
17 IQD 110.15 16 IDA 111.9

35 : S2(5255'), S4(6624'), 17 : S1(5282'), S3(6604')
34 : R2(5278'), R4(6650'), 16 : R1(5318'), R3(6706')

1st TWY GND, 2nd TWY Apron, CTC Apron

Caution HOT SPOT (HP01,HP02)

Follow Me Car on Lxx, APU off Procedure

PA

DEP **Aft T/O - Heading 190, Join W209 -> DCT**
TA98 **LATUX CRS 148(328)**

Walk Around Ramp Pass & Password 사용

3435
(350) **LAT 91D/01D** ALT 3000'(900m)

1617
(170) **LAT 81D/11D** ALT 3000'(900m)

Caution HOT SPOT (HP01, HP02)

PA

- APP 119.4 (RWY, INI ALT)



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

VTBS(BKK) 4ft

Stand F,G 1-5

KE Bangkok 131.25

UTC+7

DCL -20min, Voice 133.8

PREFLT Check Payload

- TPE - HKG RTE

ARR
TL130

SPD CTL via STAR Chart
20R/02L RNP Only

20R19
(295)

EASTE xxC

Select Tx
(No Tx)

ILS Z 19/20L
RNP 20R

02L01
(015)

EASTE xxD

Select Tx
(No Tx)

ILS Z 01/02R
RNP 02L

NAV

SVB 111.4 19 ISEN 110.5 02R ISWS 109.1
20L ISWN 109.5 01 ISES 110.1

SPD

10M' 250, 8NM 180, 5NM 150-160

19 : B8(5567'), B10(6965'), 01 : B7(5964'), B5(7962')
20L : E9(5052'), E13(7139'), 02R : E12(4872'), E7(6958')
20R : F7(6994'), F9(8471'), 02L : F6(5725'), F4(7496')
HIRO, 01/19, 02L/20R No Groov

AFT LD CONF 1, STD Taxi Route, APU OFF Procedure

PA

DEP
TA110

A-CDM REQ P/B +-5min of TSAT
TSAT/CTOT Inform to GND Staff

20L19(295)

LIPLI xJ, G/E

ALT 6000

02R01(015)

LIPLI xK, F/H

ALT 6000

APU Start within 10min, STD TAXI Route

19R Do not Pass E1, D2 01/19, 02L/20R No Grooved

- **DEP 119.25 (AUTO)**

PA

IN

Fast Track, CAPT GD, Sign 여러장

OUT

New CREW Line (에스컬레이터 밑), 여권, ID, GD
, Sign 여러장



RPLL(MNL) 75ft Gate 5,6

PAGSS Oper 131.0

UTC+8

No DATIS
DCL -5min 125.1 Voice

PREFLT 입국 세관 신고서 (<https://etravel.gov.ph/>)

- Report ETA BISIG

ARR **3 STAR Diff TRK NM (7>5>3) & IAF "RPHI"**
TL130 **GPS Interference Gate Info 11L1(L1만 사용)**

06(063) **POLIO, NABAL** GONDO ILS
7,5,3 R/P (DAGAT) RNP

24(243) **DCT MIA RDR** MUTAN ILS
Vector (MEDAM) RNP

NAV **MIA 114.4 06 IML 109.1 24 IMA 109.9**

SPD TMA 250, 20NM 210, 10NM 180, FAF 150

06 : E2(7555'), R1(8221'), 24 : E4(6725'), R5(7755')
Unable advise ATC, Do not Confuse R2, E2, RWY31

06 : Caution Vacate & RW31, 24 : TWR to APRON CTC
In Apron S4,5 Small Marking, Ask Chock in Time

PA

DEP
TA110 **Main RWY H/D Climb 7000, CLR/T/O**

06(063) **CAB xx R/A** ALT 12000/ATC

24(243) **CAB xx P/B** ALT 9000/ATC

EO

06 : MIA 063/2, MIA R250

Log on CDPLC RPHI Before DEP.

Req ENG Startup to GND -> Req P/b to Apron



- DEP 124.4
- Report ETA LEBIX

PA

IN

직원 IN/OUT GD, INFO 줌, IN GD 및 QR 제시

OUT

받은 OUT GD 제출



WSSS(SIN) 22ft

Gate G18-21

SATS FLT OPER 131.22

UTC+8

DCL -20min, Voice -5min

PREFLT

신고서 작성 (<https://eservices.ica.gov.sg/sgarrivalcard/>)

ETA KABAM 119.3 – ETA MIGUG - MAVRA 8942/5655 – MIGUG Rep
MESOX 120.7 - Rep MAPNO (RPHI – VVHM CPDLC Tx Manual)

ARR
TL130

CPDLC WSJC -10min FIR

SPD CTL via STAR Chart, No IAF RDR Vector

02LC(023)

MABAL/KARTO xA

No Tx

ILS

20RC(203)

MABAL/KARTO xB

No Tx

ILS

RNP 02R/20L : L/D CHK -> LUVUL/OBGIS -> Arm APPR

NAV

VTK 116.5 02L ICW 110.9 02C ICE 108.3
20R ICH 108.9 20C ICC 109.7 02R ICX 110.5

SPD

8NM 180kts, 4NM 150kts

02L : W5(6519'), W4(7916'), 20R : W6(5498'), W7(6423')

02C : T6(6781'), T5 (8467'), 20C : T7 (6420'), T8 (7910')

02R : A6(6738'), A5 (7995'), 20L : A7 (6250'), A8 (7395')

HIRO Unable ATC, Exit TWY 2 way but Same Name

VTK Out of A/P, TWY R/S Max 20kts, AFT LD CONF 1

PA

DEP
TA110

Req P/B +-5min TSAT

CPDLC WSJC Before DEP

Walk Around

L1 과 L2 브리지 중간에 Security가 서있는 곳으로 가면됨. 다녀
와서는 몸 수색함

20CR(203)

TOMAN xx

CHANGI xD

ALT 3000 **SPD 230 Until 4000**

02CL(023)

TOMAN xx

CHANGI xC

ALT 3000 **SPD 230 Until 4000**

PA

Standard P/B, TWY R/S Max 20kts

Cross LUSMO/AKMON xxx, T/O Before xxx

RWY 20C/R : CHK Min Climb Grad by NAVBLUE

DEP 120.3 - GUTUP (SIN 6556/8942) - HCM AKMON 133.05

- UDOSI 120.7 - ARESI 8942/5655 Unable AKOTA 119.3

IN
OUT

기계에 여권 인식하는 방식, Crew Line 따로 있음



VVDN(DAD) 30ft

Bay 14-20

Comm None

UTC+7

GND By Voice

SANYA DB Report 122.6

ARR
TL100

C/S A330 Heavy!!
SPD CTL via STAR, VOR 17 Available

35LR(353)

BUNTA xM/N/L/K
L/K : RNAV Arr Chart
Long STAR Primary

TAMLA

ILS V

17RL(173)
3.5

BUNTA xD
BUNTA xE

DAN
D0570

VOR Z
VOR Y

NAV

DAN 114.4 35R IDR 111.5 35L DAD 110.5

SPD

FL100 Max 250, 12NM Max 200, 5NM Max 160

35L : G4(6702'), G6(9803'), 17R : G2(7089'), G1(9776')
35R : E4(7198'), E6(10275'), 17L : E2(8044'), E1(10754')
NO RAPID EXIT TWY, 90 Degree TURN, AFT LD CONF 1

Follow me Car on E, E2,3,4 Narrow Over-steering

TWY D 사용 금지, 소형기 on D시만 E 사용가능

Gate IN From TWY E Only, 5m 1kts Ready for STOP

DEP
TA90

Consider Extra Fuel Due FLT ALT
ATS Plan HardCopy 제공

PA

35RL(353)

BUNTA xG/C
RNAV Deps

ALT 8000(ATC)

17LR(173)

BUNTA xA

ALT ATC

EO

35L/R : DAN 353/5, R060
17L/R : DAN/173/4.5, R130

Complete P/B – ENG Start, Follow me Car

35R CAT I Hold Point Before E1, 17L Hold Short 가까움
TWY D 사용 금지, 소형기 on D시만 E 사용가능



IN

직원이 주는 GD 제출

OUT

GD 불필요

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

WVTS(SGN) 33ft Bay 9-16

Comm None

UTC+7

Voice -15min, Del 121.8

MNL 119.3 - Monitor MNL RDO 8942(5655) - At MIGUG HCM 120.7

ARR
TL100

CPDLC VVHM

25RL(250)

PTH xD
DOVIN xD/E

SULUK
BOMSU

ILS W

07LR(070)

PTH xM
DOVIN xP/Q

SAMDU

ILS W (VOR 07L)

NAV

TSH 116.8 25R HCM 110.5 07R ITS 111.7
25L SGN 108.3

SPD

25NM 190-200kts, 5NM 150-160kts

25R : P4(6158'), P5(6991'), 07L : P3(6266'), P2(8907')

07R **DIS** :S5(6574', **110E**) **S3**(7598') then V(Heavy)

25L : S7(6824'), S8(9671') **Do not Confuse TWY S/S8**

After Vacate then TWY V

AFT LD CONF 1, Follow Me Car Service in Ramp

(Caution Sudden STOPBAR L/T)

Sensitive VDGS!!! (0.5m이내, 2m STOP시 바로 정지)

DEP
TA90

Req RWY due to Performance
Aft T/O Sudden Low ALT Maintain 2000ft

PA

Walk Around

직원이 문열어주고 기다림

25LR(250)

PATMA xA/M/P
PTH xC

ALT FL110
ALT 9000

07LR(070)

PATMA xF

ALT ATC

P/B within TSAT +- 5min, RWY CHG After TAXI

Caution Sudden STOPBAR L/T, Follow Car Service

1,2,3T is Stand name, Caution S1, P1 for Lineup

DEP 125.5

PA

IN

직원이 주는 GD 제출

OUT

호텔 CHKOUT시 GD 2장, Sign, Imm 1장, X-ray 1장



VTCC(CNX) 1036ft Bay 5-7

CNX Traffic 131.5

UTC+7

Voice -5min, GND

ARR
TL130

BKK to CNX APP 80NM, FL160 - Rep PSN,
ALT, SQ Caution TERR

36(001)

MONLO xA

PUKAT

ILS Z

18(181)

MONLO xB

PILEX

RNP

NAV

CMA 116.9

36 ICMA 109.9

SPD

20-25 TRK NM 220, IF 180, FAF 150-160

36 : E(6558', 90도), D(7483')

18 DIS : F(6853', 90도), G(9199', 90도)

Narrow RWY, No TWY CL L/T, Yellow Center Line

Difference VDGS, AFT LD CONF 1

PA

DEP
TA110

AFT T/O L/H TURN DCT MONLO then R-207
CHK CLB Gradient, RECMD FLEX Due EGT Limit

36(001)

MONLO xN

ALT 11000

18(181)

MONLO xS

ALT 6000

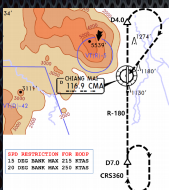
EO

36 : CMA 001/4, R180

18 : CMA 181/7

DEP 129.6

PA



IN

BKK 처럼 CAPT GD Sign 5번, Gate 7번 앞 버스 대기

OUT

GD Sign 5번



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

WADD(DPS) 14ft Stand A15-25

Gapura Oper Bali 131.95 UTC+8 Voice -20min, TWR/GND

PREFLT QR code 작성 (<https://allindonesia.imigrasi.go.id>)

CPDLC - POTIB 10-25 min RPHI - MAMOK 15-30 min WAAF

ARR TL130 RW27 Before DOBMA(EMTAS) FL300 for CTC Bali 119.7 (Prevent Late DES)
Late DES due TERR

09(087) DOBMA/EMTAS xC KHUTA RNP

27(267) DOBMA/EMTAS xD BENOA ILS

NAV BLI 116.2 27 IDPS 110.3

SPD 10NM 180kts, 4NM 150kts

09 : N3(6817', 90도), N2(8448', 90도)

27 : N6(7513', Rapid), N7(9645', 90도)

Narrow RWY, No RWY/TWY CL L/T, Yellow C/L Only

VDGS but Marshaller, GND REG No, After LD CONF 1

PA

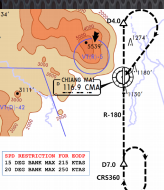
DEP TA110 +/-15min Req P/B, Rep Bay, Resist No, POB
CPDLC Before DEP WAAF

09(087) DOBMA/EMTAS xA ALT 9000

27(267) DOBMA/EMTAS xB ALT FL240(ATC)

EO 09 : BLI 087/6, R130

Caution Intermediate HP W/E



DEP 119.7
CPDLC

- Before DEP WAAF - GORAI 10-25min RPHI
- GURAG RJJJ

PA

IN 도착후 직원이 주는 입/출국 카드 작성, GD 불요, QR 제시

OUT KE 카운터 앞 Dispatch 브리핑 후 인솔, Imm 이후 STAFF Only로



VDTI(KTI) 22ft

Bay 107-123

Techo DIS 129.0

UTC+7

Voice P/B -10min, GND

CPDLC- POTIB 10-25min RPHI, MONBO 15-45min VWHM
ETA KABAM(RTA) – ETA MIGUG - Report MAVRA, Report MIGUG
At MIGUG CTC HCM 120.7

ARR
TL ATIS

Do not Confuse ANAT, ANAB
TCAS RA Des Rate!!

05(048)

SUSEL/ELNAK xD

NULPI/IKOMO

ILS Z

23(228)

SUSEL/ELNAK xC

AKPEG/LELUN

ILS Z

NAV

TIA 111.0

23 IKTI 110.1

05 IKTA 108.75

05 : A5(6562'), A6(7677'), 23 : A4(6562'), A3(7677')
Rapid Exit, Non-Grooved RWY

Caution Complex Lead Line into MARS

PA

DEP
TA100

RWY 23 Do not Overshoot B17

05(048)

ELNAK x

ALT ATC (6000)

23(228)

SUSEL x

ALT ATC (6000)

APP 123.8 (AUTO)

CPDLC

- AKOTA 15-45min VWHM - POTIB 10-25min RPHI

ETA ARESI - MNL RDO 8942/5655 At ARESI

MNL 119.3 At -10NM AKOTA - ETA POTIB

PA

IN

GD 2장, 입국, 출국 각각 사용

OUT

출국장 맨 왼쪽 CREW LINE



RCTP(TPE) 108ft

Bay A1-9

Dynasty Oer 131.3

UTC+8

DCL Avail, Voice -5min

ARR
TL 130

COPRA RL200

05LR(054)

BAKER xA

JAMMY

ILS

23RL(234)

BAKER xB

AUGUR

ILS

SPD

FL250-130 280, FL130-10M' Max 280 Ref APP

NAV

no VOR 05L ITIA 111.1 23R ITYA 109.3
05R ICKS 110.7 23L ICJN 111.9

05L : N7(5787'), N8(7411'), 23R : N4(6656'), N3(9340')
05R DIS :S6(5419'),S7(7244'), 23L DIS :S5(5442'),S4(7470')

Center Green, Deviation by Arrow, Distance Yellow



PA

DEP
TA110

Be Ready Intersection T/O
In DCL "MNTN A030" -> 3000ft

05RL(054)

PIANO xA/C

ALT ATC

23LR(234)

PIANO xD/B

ALT ATC

"DCT PIANO then L3 RNAV Transition"

PA



YYYY(RGN) 110ft **Stand 6,7**

Oscar Papa 131.8

UTC+6:30

Voice -15min, GND

PREFLT 승무원 2 SET 운영, 탑승 정비사

NAR(No Alternate Airport) not apply due Single RWY

ARR BGO(BAGOO), HGU(HLEGU) **ILS Z GNSS Req**
TL 075 **GPS Interference -> RDR Vector + ILS Y 21**

21(214) **GPS OUT** **RDR Vector** **ILS Y**
RELAP xA **PAKSU** **ILS Z**

03(034) **BAGOO xB** **KOSDA** **RNP**

NAV **no A/P VOR BGO HGU 21 IYGN 109.9**

21 : A3(8152'), R1(8415'), 03 : A5(7972'), A6(11069')
RWY 21 Upslope(0.6%), Exit End of RWY or By ATC

Caution WingTip (Watch out Safety Line)

Ramp Area Dark(Consider LD L/T)

PA

DEP **GPS 1+2 Fail MEL34-50-08F, 34-50-09B(RTG)**
TA60 **No RNP RTE, ADS-B No RVSM Req Equip ->**
RADIONAV Updaate within 1NM No Problem

SEL **RDR Vector** **SEL HD 214 ALT ATC**
RW21 **(VISAM xA/C)** **SID GNSS Req**

SEL **RDR Vector** **SEL HD 034 ALT ATC**
RW21 **(BAGOO xA)** **SID GNSS Req**

RWY 03 Taxi Upslope(more PWR)

RGN RDR 128.75

- After BOMAS GPS Back to NOR
- Sometimes Spoofing

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

WMKK(KUL) 69ft Stand C2,4,12

Layang Sat. 132.9

UTC+8

Voice TSAT -10min

ETA KABAM 119.3 – ETA MIGUG MAVRA 8942/5655 – MIGUG Rep

MESOX 120.7 - Rep MAPNO (RPHI – VVHM – WSJC - WMFC)

ARR
TL130

ATC Req POB
Long TRK Mile STAR Expect Short Cut

33R33(326)

SAROX xG/H

EGURI

ILS

14L15(146)

+ **Initial APP Chart**

MESUP

SPD

20NM 220, 15NM 185, 10-5NM 160

NAV

VKL 116.1 33L IWL 111.9 33 IWM 111.5
14R IWR 110.7 15 IWK 110.1

32R : A5(6938'), A4(8343'), 14L : A6(6922'), A7(8425')
33: Y5(6948'), Y4(8425'), 15 : Y6(6948'), Y7(8415')

RWY 32L via C5 Caution D5,6

Wingtip CLR P1-3, Q5-7, Y1-8 ATC Req HLxx/LD Time

Standard TWY RTE, AFT LD CONF 1

PA

DEP
TA110

Req P/B +-5min of TSAT
CTOT from Staff

32L33(326)

KIMAT xE/F

ALT ATC(6000)

14R(146)

KIMAT xB
RNP 1600-2100Z

ALT ATC(6000)

Standard TAXI RTE, P/B Exit S8 -> Taxi out S8
Cross DUDIS xxx, T/O Before xxx

DEP 135.25(Before 2000ft)

PA

GUTUP 6556/8942 – AKMON 133.05 - UDOSI 120.7

ARESI 8942/5655 Unable AKOTA 119.3

IN 브리지 밖 직원 GD, 2300L 이전 Aerotrains, 이후 BUS(C5 Sign)

OUT 호텔 GD 수령, Aerotrains to Gate



WVCR(CXR) 46ft Bay 36

Comm None

UTC+7

Voice TWR 118.2

ETA KABAM 119.3 – ETA MIGUG MAVRA 8942/5655 – MIGUG Rep
MESOX 120.7 - Rep MAPNO (RPHI – VVHM)

ARR
TL100

Wx, RWY from HCM, **CHK LDW due to PCR**
RWY20 Max Tail wind 15kts
CAAV STAR, APP not Authorized

20LR
(200)

COTUN, BANKE,
HUNTA, NHATA xx

CRxx
AFT IAF
STEEP
DES(3.8)

ILS Y 20L
RNP 20R

02RL
(020)

ILS X/Z

NAV

CRA 116.5 20L ICA 110.3 02R ICR 111.9
02L ICR 110.7

20L : G3(6735'), G1(9603'), 02R : G5(6528'), G7(9662')
20R : W4(5971'), W3(7680'), 02L : W5(5606'), W6(7345')

FollowMe Car Service, **Sensitive VDGS Caution**

PA

DEP
TA90

CHK TOW due to PCR
Follow Restriction due to Military Traffic

02LR(020)

NiHOA xA

ALT 7000

20RL(200)

NiHOA xB

ALT ATC(FL100) SPD 250

EO

02 : CRA 020/3, R090, 20 : CRA 200/6, R150

TWY Y5 only below wingspan 36m/118ft

DEP 127.9

HKG RTE

PA

IN
OUT

여권 제시 특이사항 없음



WVPQ(PQC) 23ft

Bay 7,8,11,13A

Comm None

UTC+7

Voice GND 121.92

HCM 120.9 (STAR,RWY,Wx) – No APP Freq

At OSOTA PQC TWR 118.6

ARR
TL100

CPDLC VVHM
STAR RWY Wx from PHN or HCM(120.9)

28(276)

OSOTA xE
TRUMI xC

PQ328
TUNPO Tx, PQ428

ILS Z

10(096)

OSOTA xF
TRUMI xB

PQ310
TUNPO Tx, PQ210

ILS Z

NAV

PQU 113.0

28 IPH 108.7 10 IPQ 108.1

28 : S6(6758'), S8(9662'), 10 : S4(6761'), S2(9665')

RWY 28 PAPI R/H not Accurate

AFT LD CONF 1, Stand 13A only Widebody
(Caution Sudden STOPBAR L/T)

Sensitie VDGS!!! (0.5m이내, 2m STOP시 바로 정지)

PA

DEP
TA90

Req RWY Due to Performance
AFT T/O Sudden L/O 2000ft

28(276)

OSOTA xC

ALT 6000 SPD 250

10(096)

OSOTA xD

ALT ATC

LINE UP via RWY 28 S2/S1 or RWY 10 S8/S9

TWR 118.6

- Before FL100 PNH 132.4/124.7 or HCM 120.9

PA

IN

GD 2장, IMM 1장, 세관 1장(세관은 안받음)

OUT

KE 카운터 GD 2장, IMM 1장, X-ray 1장



VVNB(HAN) 43ft

Bay 15-21

Comm None

UTC+7

Voice -15min GND

SANYA BD Report 122.6

ARR
TL100

TEBAK RL240, SPD CTL via STAR Chart
GS Fluctuation

11LR
(109)

NAKHA xH/E/F/G
J/K : RNAV ARR

VILOX
NB362

ILS X

29RL
(289)

NAKHA xL/M/N/P
RNAV ARR

NB462
NB452

ILS X

SPD

25NM 190-200, 10NM 180-160, 5NM 15-160
Max RWY 40, TWY 30, Apron 15, Turn 10

NAV

NOB 116.1 11L NB 110.5 29R INA 109.3
11R NBA 108.3 29L INB 111.9

11L : **P7(6929')**, **P8(8366,90도)**, 29R : **P4(6361')**, **EOR**
11R : S7(7555'), S8(10377'), 29L : S4(7732'), S3(10213')

AFT LD CONF 1, Follow me Car on S
5m 1kts Ready for STOP

PA

DEP
TA90

Consider Extra Fuel Due CRZ Level
TEBAK FL250

11RL(109)

NAKHA xA
CHK NB110

ALT ATC

29LR(289)

NAKHA xC

ALT ATC

STD TAXI RTE - "DEP 1B"

TWY S1,3,5,6,8 항공기 있으면 TWY S 금지

DEP 125.1

HKG RTE

PA

IN

GD 불요, IMM 좌측 CREW LINE

OUT

DEP B(둘중 오른쪽)



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

YBBN(BNE) 15ft

Stand 78-85

Menzies BNE 132.55

UTC+10

DCL -5min Voice

PREFLT

CREW Declaration 작성

CPDLC & PSN Report : RJJJ – KZAK(FIR) –AYPM(FIR)-YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

FIR PSN Report Caution Wx DEV Sequence

ARR

TL110

Check ALT A/P VBCG L/D WT

SPD Restriction in Chart + PRM

01L/R(016)

TANIK/GLENN

ILS

SMOKA xA

19R/L(196)

OSOKO/ESTSO

ILS

NAV

BN 113.2 01L IBN 111.5 19R IBE 109.3

01R IBA 109.5 19L IBS 110.1

SPD

5NM 150-160kts

01L : T6(6243'), **T4(7598')**, 19R : T9(6246'), **T11(7595')**

01R : **A4(7043')**, **A3(8579')**, 19L : A6(5620'), **A7(8477')**

Prefer Exit Unable ATC, min REV TH, T5, T10 No Exit

AFT LD CONF 1, CTC Apron Before Entering

Report Cockin Time Caution Hotspot RWY 31

PA

DEP

TA100

CPDLC YBBB on GND 45min/Passing 10M'

**"CNL SID, TRK EXT CL, xx Degree" -> Main
RWY TRK**

01L/R(016)

BIXAD x 01R Diff Chart

ALT ATC(6000)

19R/L(196)

BNE x RDR SID with H/D

ALT ATC(6000)

PushBack Point C,D,E , TWY A for T/O

Do Not PASS B1 for 19L

- DEP 133.45

PA

IN

로툰다 나온 직후 KIOSK 사용(한산함)

OUT

DEP 입구옆 직원라인 진입

EDTO

Next Page



EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning (Ops Specs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft**

(PFD/ISIS 60ft, ST-BY 300ft)

NAV DATA Input : **EEP, ETP1, ETP2, EXP**

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Threshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : **EEP**, FIX 2 : **ETP1**

FMS ALT A/P SET : **DATA – ETP Page(Nearest AP)**

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Review Contingency Procedure

- **Drift Down 30도이상, 5NM, FL290이하, +-500ft**

- **Wx Dev 5NM 이상, +-300ft**

EDTO Segment

Apply Actual Wx for Actual Divert

TERR Escape RTE Papua New Guinea

COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

Diversion

- **Enroute RNP downgrade**

- **Weather min ERA AP**

- **LAND ASAP by ECAM, QRH**

- **Only 1 GEN(IDG, APU, EMER) + Multi Fail**

- **Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)**

- **Increased Fuel Consumption**



ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Last ETP(Critical Point) Fuel less than PLAN – **Continue by PIC**

EXP (Exit Point)

YSSY(SYD) 21ft

Stand 78-85

Menzies SYD 129.6 UTC+10/11

Voice CLR 133.8/GND 121.7

No DATIS

PREFLT

여권, Custom Form

CPDLC & PSN Report : RJJJ – KZAK(FIR) –AYPM(FIR)-YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

FIR PSN Report Caution Wx DEV Sequence

ARR
TL110

RW34L/16R on REQ

CHK SEMAP, Curfew 2300-0600L

ILS PRM, NADP Proc Review

16LR(155)

BOREE xA

URDEN

ILS

34RL(335)

(xP for 16R)

SOSIJ

ILS

NAV

SY 112.1 34L ISN 110.1 16R IKS 109.5

34R IKN 109.3 16L ISS 110.9

SPD

10NM 185-160, 5NM 160-150

34R : **U1(5515')**, L(7378'), 34L : **A3(6207')**, **A2(6620')**

16L **DIS:T4(5718')**, T6(6971'), 16R : **A3(6210')**, **A4(6561')**

Prefer Exit Unable ATC, min REV TH, NO LASHO

SY 20NM Below 9000ft, Intl x, Dom x - TWY

Same Spot no. International/Domestic

PA

DEP
TA100

CPDLC YMMM on GND 45min/Passing 10M'

CASA no Approval KAL (for T/O min)

16RL(155)

GROOK x RIC tx
(SYD x RDR)

ALT ATC SPD 270

34LR(335)

RIC tx
(SYD x RDR)

ALT ATC

CHK Special Requirement SID Chart

- **DEP 123.0**

PA

IN

OUT

EDTO

Next Page



EDTO Procedure

PREFLIGHT

Apply Alternate Airport IFR Wx Minima for Planning (Ops Specs C055) -> EDTO ERA Only(ERA no Wx)

RVSM CHK : **CAPT/FO 20ft, PILOT/FE 75ft**

(PFD/ISIS 60ft, ST-BY 300ft)

NAV DATA Input : **EEP, ETP1, ETP2, EXP**

HF SELCAL CHK : Jeppesen - ENT DATA Pacific

SEOUL RADIO : 8903(3004,6532,13300,13303,17904)

AFTER LEVEL OFF (CRZ CHK)

RVSM CHK : **CAPT/FO 200ft**

BEFORE EEP (Entry Point, ERA 기준)

60min(Threshold) : A330 420NM

One Eng : CRZ SPD 310kts, 180min 1235NM

FIX 1 : **EEP**, FIX 2 : **ETP1**

FMS ALT A/P SET : **DATA – ETP Page(Nearest AP)**

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Review Contingency Procedure

- **Drift Down 30도이상, 5NM, FL290이하, +-500ft**

- **Wx Dev 5NM 이상, +-300ft**

EDTO Segment

Apply Actual Wx for Actual Divert

TERR Escape RTE Papua New Guinea

COMM (CPDLC)

RJJJ – KZAK(FIR) – AYPM(FIR) – YBBB(FIR)

(Cross GUM airspace 250NM 118.7, SQ2100)

Diversion

- **Enroute RNP downgrade**

- **Weather min ERA AP**

- **LAND ASAP by ECAM, QRH**

- **Only 1 GEN(IDG, APU, EMER) + Multi Fail**

- **Only 1 Main GEN(IDG, APU) + G HYD(low LVL, PRESS, Overheat)**

- **Increased Fuel Consumption**



ETP (Equal Time Point, EDTO ERA기준)

FIX Change, ETP Page SET

EDTO C/L : **Fuel, A/C, MSA, ALT Wx & NOTAM**

Last ETP(Critical Point) Fuel less than PLAN – **Continue by PIC**

EXP (Exit Point)

VNKT(KTM) 4395ft Spot 1-4

Nepal Airline Corp 131.6 UTC+5+45

No DATIS
Boarding ATC CLR

CPDLC VYYF(Yangon), VECF(Kolkata)

Kolkata -10min (125.77/133.75) - Dhaka -10min ADC No. (125.7)

- KTM -10min (126.5) C/S, A333, PSN, FL, FIR Time RTE, L/D type

"Req Entering CLR"- Report Kolkatka Establish with KTM

ARR **Curfew 1915-0015Z**

TL150 **Night : Only RNP Z 02(AR) Comm Policy**

02(022) **ROMEO xR** RATAN RNP Z 02 / VOR 02

SETUP **Nav : KTM 105&288, Fix : KTM /4, DANFE**
FMS : KTM530/D9.0 (Vapp)

NAV **KTM 113.2 02 IKT 108.1**

02 : **B(6814')**, A(8395'), 20 : E(**6387', 90°**), 180 Back
High FE + Upslope = 50' Flare (No 40', 30')

RAMP SLIPPERY NO Release Brake

PA

DEP **Req ADC & FIC, P/B with POB, Intersection A, E**
TA135 **If Wx is Severe, Stand-by at GATE**

20(202) **IGRIS xA** ALT ATC

02(022) **IGRIS xB** ALT ATC

EO 20 : KTM /4, R340, R288 02: KTM /3, R288
Maintain CONF Until 10500ft, 8000ft LVR MCT(10min)

RW20 Takeoff at Painted Threshold

D2.5 KTM/KTM Turn Even Below 400ft (Max 180kts)

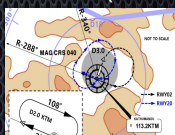
DEP 120.6 CPDLC VECF, VYYF

- Dhaka -10min ADC No. (125.7)- Kolkata -10min (125.77/133.75)

Radio Problem

- -80NM KTM FIR to Kolkata FIR +30NM

KMG 125.75(15-20min FIR)



PA

IN

OUT

RNP AR
Next Page



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

A330 KTM RNP Z 02(AR)

Before TOD: **STATUS, Review VOR 02, Temp -10°C**

Must GA
- V/DEV 75ft
- D/DEV 0.3NM
- SYS Malfunction

L/D CONF
Before L/D C/L
FINAL APP
GA ALT 10500' set

Both ALT within 100ft
TERR ND 10NM
PF-F PLN / PM-PROG
RNP 0.3NM PROG

Flap 1

A330 KTM VOR 02

Do not use FINAL APP

At D3.0 FPA -3
Visual L/D

Before D9.0
L/D CONF
Before L/D C/L
SPD Vapp

8900ft ALT HOLD
D9.3 set GA ALT
FPA -5.3 Set & Pull
Monitor Pass ALT
S/B Needed

FLAP 1

SYS Fail / GA (Dual-GA, Single-GA or CONT)

if Follow GA RTE (Acc HT 10500ft)

- Reach 10500ft – **TH CLB - Acc** – Flaps up – Req DANFE

Dual FMGC (NO GA RTE) (Acc HT 10500ft)

- Man GA, 15°, 10-20kts ADD (**At RW02 R/H Turn**)

- PM **NAV B/UP** both **MCDU Menu** – TRK/FRA – Follow TRK (A/P ON)

- **if No GA RTE Follow VOR GA Procedure**

- Reach 10500ft – **TH CLB - Acc** - Flaps-up – Req DANFE

VOR GA Procedure (Acc HT 10500ft)

At RW02 R/H Turn (**Max 185kts**) – 4 DME Arc(IN 5 DME)

After RW02 Track 288 OUTBD Until Reach 10500ft

ZMCK(UBN) 4485ft Spot 2,4,6

Comm None

UTC+8

No DATIS
Voice TWR 129.0

5min Before POLHO

Boundary Report to UBN 126.5

ARR
TL148

HI FE, TERR/Restrict Area

CTC UBT 5min Before FIR(POLHO, INTIK)

HI FE&Sink Rate due TAS

29(290)

MENUG xW

ZONAM

ILS X

11(110)

MENUG xV

CK113

RNP(Req)

NAV

SER 116.9 29 CDA 109.9

29 : C(5761'), **B(8241')**, **END RWY**, 11 : C(5761'), D(8231')

Gate 2 A330 Must Follow Dashed Line

PA

DEP
TA128

Nothing Special

29(290)

MENUG xA/C

ALT ATC

11(110)

MENUG xB/D

ALT ATC

EO

29 : SER 290/4, R343, SER R133, SER 290/10, R260

11 : SER 110/1.5, SER R313, SER 290/10, R260

TWY F very Narrow, TWY B,D Opposite Turn Prohibit



DEP 120.0 UBN 126.0 - 126.5

Boundary Report 15-20min

PA



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

OMDB(DXB) 62ft

Stand C45-63

DNATA Oper 131.6

UTC+4

DCL Avail, Voice -10min

S RTE : CHILA Report YANGON, DAKHA, KOLKATA

Boundary Report - PURPA(W112) : islamabad Before 15min

- APELO(B505) : Karachi Before 15min- DERBO(G452) : Air Defense

RDR Before10min - over IRAN : Air Defense RDR (127.8/135.1)

PSN+FL+DEP+DEST (Separate Tehran CTL)

ARR

TL150

Arr : C/S,A333, Passing FL -> Director : C/S Only, SPD Restriction in JEPP

12LR(119)

PUVAL, IMPED xB

REREK

ILS

30LR(299)

PUVAL, IMPED xC

ULDOT

ILS

NAV

12L IDBL 110.1 30L IDBW 111.3

12R IDBE 109.5 30R IDBR 110.9

12L : M7A(6135'), M9(7253'), 30L : K9(6811'), K8(8041')

Late L/D CLR, Assigned Rapid Exit TWY

TWY J,J1,J2,J3,U,W,Y,Z Max 15kts, VDGS 2kts

Caution TWY Holding Point, No Wing Walker

PA

DEP

TA130

Full Length to ATC

30R(299)

DAVMO, IVURO xF

ALT 4000 SPD 220

12R(119)

DAVMO, IVURO xG

ALT 4000 SPD 220

DEP 124.675 C/S, Passing LVL, Unable SPD

CTC Air Defense RDR 127.8/135.1 Before 10min Tehran FIR(GABKO FL190) - MUM HF : 6661/8879/5658

S RTE : CHILA Report YANGON, DAKHA, KOLKATA

Boundary Report - KMG 125.75(15-20min FIR)

PA

IN	GD 2장 - IMM 1장, OUT시 1장 (객실꺼 따로) - 객실과 STAY 다름
OUT	6번 맨좌측 Crew/Staff Sign - 출국 GD+티켓+짐부침 - IMM IN/OUT GD 두장 제출(객실 따로)



Meter/Feet Conversion Table			
Westbound		Eastbound	
		13,700 m	44,900 ft
13,100 m	43,000 ft	12,500 m	41,100 ft
12,200 m	40,100 ft	11,900 m	39,100 ft
11,600 m	38,100 ft	11,300 m	37,100 ft
11,000 m	36,100 ft	10,700 m	35,100 ft
10,400 m	34,100 ft	10,100 m	33,100 ft
9,800 m	32,100 ft	9,500 m	31,100 ft
9,200 m	30,100 ft	8,900 m	29,100 ft
8,400 m	27,600 ft	8,100 m	26,600 ft
7,800 m	25,600 ft	7,500 m	24,600 ft
7,200 m	23,600 ft	6,900 m	22,600 ft
6,600 m	21,700 ft	6,300 m	20,700 ft
6,000 m	19,700 ft	5,700 m	18,700 ft
5,400 m	17,700 ft	5,100 m	16,700 ft
4,800 m	15,700 ft	4,500 m	14,800 ft
4,200 m	13,800 ft	3,900 m	12,800 ft
3,600 m	11,800 ft	3,300 m	10,800 ft
3,000 m	9,800 ft	2,700 m	8,900 ft
2,400 m	7,900 ft	2,100 m	6,900 ft
1,800 m	5,900 ft	1,500 m	4,900 ft
1,200 m	3,900 ft		
ALT/HEIGHT Conversion			
1,000 m	3,300 ft	500 m	1,600 ft
900 m	3,000 ft	450 m	1,500 ft
800 m	2,600 ft	400 m	1,300 ft
700 m	2,300 ft	350 m	1,100 ft
600 m	2,000 ft	300 m	1,000 ft
550 m	1,800 ft		

Difference (ATS A333)

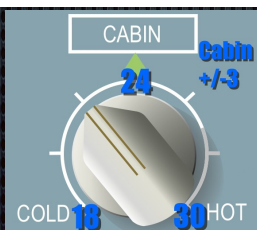
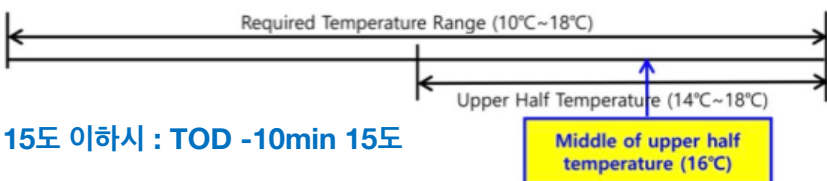
333Q(3)		HL7753, 7754, 7701 C24+Y252(151)	
333R(8)		HL7584-7720 C24+Y260(156)	PACK LO Y Class 60%이하
333R(5) (OZ)	APMS	HL7740-7754 C30+Y260(156)	PACK NOR All Class 200이상
333R(9) (OZ)		HL7792-8293 C30+Y268(160)	
333E(6)	APMS -1.5	HL8001-8027 C24+Y248 HL8001 RNP AR X	RNP AR, OANS, PAX AUTO, TRIM AUTO

ENG START : Basic 43-48% Enhanced 52-54%
ENG OIL : 12Qt + 0.5Qt/h

Categories

WING SPAN	60.30m	197.83ft
180 Back (72도)	48m	157ft
WING SPAN CAT	ICAO E	FAA V
Wt CAT	Heavy	
APP CAT	ST-IN : C	Circling : C/D(FAA)
524700lbs	238T	507063lbs 230T
478402lbs	217T	436515lbs 188T
423287lbs	192T	

LOADSHEET : CARGO No. FWD 11-21, AFT 31-41



CRZ FUEL Penalty (Approximation)

A330

4000ft below OPT ALT : 5% increase trip fuel

8000ft below OPT ALT : 10% increase trip fuel

Above OPT ALT : 3% increase trip fuel

FUEL Consumption

APU

GND : 438 LBS/HR (BLOCK FUEL산정)

IN FLT : 286 LBS/HR

TAXI

2 ENG, no APU : 3300 LBS/HR (10분 550 LBS)

CRZ

100FT 당 8 MIN

ex) 1000FT 상승 80 MIN, 1HR 20MIN

Holding

분당 200LBS

(Actual F/F 참조, 시간당 12000LBS)

GA &

3000 LBS

Landing

(과거 EDTO자료 1회 Missed App+L/D)

FUEL Loading

Tail Tank :

CL TANK 80000LBS 이상 5000LBS 까지

CL TANK 160000LBS 이상 10000 LBS 까지

FUEL Overfill : 1000LBS 기준



ONE ENG TAXI AT ARRIVAL

Taxi with engine 1 is recommended in order to pressurize the blue hydraulic system to ensure brake accumulator pressure

Note: If taxi is performed with engine 2, the flight crew must check that the brake accumulator pressure on the BRAKES and ACCU PRESS indicator is normal.

AT ARRIVAL

APU..... START / CHECK AVAIL F/O

- After high thrust operations:

ENG MIN COOL TIME ...CONSIDER CAPT

Note: The flight crew should operate the engines at or near idle thrust for a cooling period of 5 min before engine shutdown, in order to thermally stabilize the engines.

- When taxiing in a straight line:

ENG 2 SHUTDOWNORDER CAPT

Note: During engine shutdown, a slight jerk forward may occur if the brakes are applied while the aircraft is moving.

ENG 2 ANTI ICE pb-sw OFF F/O

ENG 2 MASTER lever..... OFF F/O

ENG 2 PARAMETERSCHECK F/O

X-BLEED OPEN F/O

(To supply both packs from ENG 1)

- At parking:

ENG 1 SHUT DOWN CAPT



PACKS USE WITH LP AIR CONDITION UNIT

(SD 연결 표시없음)

공항 요구로 APU OFF후 기내 온도 조절을 위한 방법
Low Press Unit을 연결하며 APU와 동시사용 금지

The flight crew must not use conditioned air from the packs and from the LP Air Conditioning Unit at the same time, to prevent any adverse effect on the Air Conditioning system.

절차 POM, FCOM에 없음

APU BLEED OFF 후 LP AIR 연결

EXT PWR 연결후 APU OFF (AVAIL -> ON)

APU Start - LP Disconnect - APU BLEED ON

APU BLEED USE WITH HP AIR START UNIT

(SD 연결 표시없음)

APU 부작동시 AIR START UNIT으로 시동을 위해 사용
외부 BLEED AIR의 역할을 함. APU BLEED 동시사용 금지

The flight crew must not use bleed air from the APU BLEED and from the HP Air Start Unit at the same time, to prevent any adverse effect on the Bleed Air

• Before connecting the air start unit:

PACK 1 & 2 as REQ F/O

HP AIR 시동전 Air Condition으로 사용시 ON (시동시 OFF)

APU BLEED OFF F/O

ENG 1 & 2 BLEED OFF F/O

X BLEED OPEN F/O

AIR START UNIT CONNECT ... REQ CAP

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.



ENG START
Next Page

STARTING with AIR START UNIT

통상 ENG 1 먼저 (EXT PWR Start 2 ENG)

“Req Engine Start up Present Positon~~~”

- Before connecting the air start unit:

PACK 1 & 2 OFF F/O

prevent any possible contamination of the packs by the air start unit.

OFF L/T 들어와야함.

APU BLEED OFF F/O

ENG 1 & 2 BLEED OFF F/O

X BLEED OPEN F/O

AIR START UNIT CONNECT ... REQ CAP

Two ground air start units may be used in parallel if the pressure/flow relation is expected to be marginal.

- When cleared to start:

Note **AIR ABNORM BLEED CONFIG** alert is triggered after the first engine start. It can be disregarded.

ENG 1 START CAP

Note: As necessary, engine 2 can be started first.

In this case, check the brake ACCU pressure prior to engine start.

Note: The minimum recommended starter air supply pressure is 25 PSI when the start valve is open.

- After engine 1 is started: **AVOD 다 꺼짐**

- If the air start unit is used to start engine 2:

ENG 2 START CAP

- When engine 2 is started:

AIR START UNIT DISC . . . REQ CAP

X BLEED AUTO F/O

ENG 1 & 2 BLEED ON F/O

PACK 1 & 2 ON F/O

- If the crossbleed engine start procedure is used to start engine 2: **EXT PWR - AVAIL first!!**

AIR START UNIT(EXT PWR)DISC.RQ ... CAP

ENG 1 BLEED ON F/O

PACK 1 & 2 ON F/O

CRSBLD ENG START PROC.APPLY

EXT PWR CHK AVAIL CAP

EXT PWR DISCONNECT REQ CAP

NOTE : EXT PWR removed after 1 ENG start.

but **FCOM :** In order to ensure that the entire network remains electrically supplied, it is recommended to start both engines before the disconnection of the external electrical power.

NOR PROC - AFTER STARTRESUME



**CrossBleed Start
next page**

ENG CROSSBLEED START

#1 ENG BLEED 로 #2 ENG START

Req PushBack

PushBack 완료(Parking Brake set)

One engine must be running in order to supply air for other engine start.

- Before second engine start :

APU BLEED OFF F/O

The BLEED valve of the supplying engine reopens and the cross bleed valve closes.

ENG BLEED (supply engine) ON F/O

ENG BLEED (receive engine) OFF F/O

The bleed valves of receiving engines are closed to avoid reverse flow leakage.

X BLEED..... OPEN F/O

- When cleared to start :

AREA CLEAR OF OBS CONF CAP

Ensure increased power jet wake does not constitute any hazard to people or installations behind the aircraft.

THR LVR (sup eng) . ADJ BLD PRES CAP

Adjust thrust of supplying engine to obtain an engine bleed pressure of **30 PSI**. If the thrust required to obtain the appropriate engine bleed pressure exceeds **40 % N1**, pay particular attention to the surrounding area.

RECEIVING ENGINE START ALL

Apply the normal engine start procedure.

- After start :

THR LVR (supply engine) IDLE CAP

X BLEED..... AUTO F/O

ENG BLEED (receive engine) ON F/O

PACK 1 & 2 ON F/O



COLD TEMP CORRECTION General

5도 간격은 보수적으로 보간법 적용됨

Min 제외한 모든 고도 수정은 ATC 인가 필요
Mandatory, Missed App 고도 ATC 사전 인가 없이 금지
반드시 고도 - FE 후의 고도를 보정해야함.
Ex) FE 200ft 공항 : 5000ft는 4800ft만 보정해야함.

Height Above FE (Feet) 200-800ft							
TEMP	200	300	400	500	600	700	800
0	20	20	30	30	40	40	50
-5	20	30	40	40	50	60	70
-10	20	30	40	50	60	70	80
-15	30	40	50	60	80	90	100
-20	30	50	60	70	90	100	120

Height Above FE (Feet) 900-5000ft							
TEMP	900	1000	1500	2000	3000	4000	5000
0	50	60	90	120	170	230	280
-5	70	80	120	160	230	310	390
-10	90	100	150	200	290	390	490
-15	110	120	180	240	360	480	600
-20	130	140	210	280	420	570	710

GMP, CJU, PUS next page



COLD TEMP CORRECTION (Domestic)

Min 은 반드시 수정 (중간 고도 CORRECTION은 PIC 결정)

Missed App 고도는 ATC 협조 필요

GMP 32L (261') / 32R (262') / 14R (254')

32L/R	8000	5500	5300	4000	2800	2300	2000
0	8450	5810	5600	4230	2970	2440	2120
-5	8620	5930	5710	4310	3030	2490	2160
-10	8780	6040	5820	4390	3080	2530	2200
R14	4000	2800	1400		4000		
0	4230	2970	1490		4230		
-5	4310	3030	1520		4310		
-10	4390	3080	1540		4390		

CJU 07 (307') / 25 (296')

	4000	2900	1800	07	8000	25	6000
0	4220	3070	1900		8450		6340
-5	4300	3130	1940		8620		6460
-10	4380	3180	1970		8780		6590

PUS 36L(233'),36R(228') / 18L/R (see below)

36L/R	6000	5000	3300	2100		6000	
0	6340	5290	3490	2210		6340	
-5	6460	5390	3560	2250		6460	
-10	6580	5490	3620	2290		6580	
18L/R	6000	5000	4000	2600	1700		6000
0	6340	5290	4230	2760	1800		6340
-5	6460	5390	4310	2810	1830		6460
-10	6580	5490	4390	2860	1870		6580



COLD Wx Operation 1/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PRELIMINARY (OAT 3°C 이하)

PROBE/WIN HEAT – ON (ENG START후 AUTO)

LOGBOOK ----- CHECK

MAX TIME TO NEXT ENG ACC ---- DETERMINE

TAXI IN TIME – NEXT ENG ACC TIME 15분 이내인지 확인

TAXI-OUT(OAT 3°C 이하)

TAXI TIME ----- MONITOR

15분 초과 또는 Eng Vibrations시 절차 수행

SOP SURFACE COND & AREA ----- CHECK

ATC ----- NOTIFY

PARKING BRAKE ----- ON

the aircraft starts to move, immediately retard the thrust levers to IDLE.

THR LEVERS ----- 50% N1 (NO HOLD TIME)

THR LEVERS ----- RETARD TO IDLE

Repeat this procedure at intervals not longer than 15 min, or if the engine vibrations increase.

SOP TAXI ----- RESUME

BEFORE TAKEOFF

FLAPS lever ----- SET FOR TAKEOFF

FLAPS ----- CHECK PSN

T/O CONFIG pb ----- TEST

T/O MEMO ----- CHECK NO BLUE

AFTER START checklist ----- COMPLETE

BEFORE TAKEOFF checklist --- ACCOMPLISH

TAKEOFF(OAT 3°C 이하)

A final engine acceleration must be performed before takeoff to entirely clean the engine from ice accretion.

ATC ----- NOTIFY

PARKING BRK(OR BRK PEDALS) ----- APPLY

THR LEVERS ----- 50% N1 (NO HOLD TIME)

THR LEVERS ----- RETARD TO IDLE

SOP TAKEOFF ----- RESUME

AFTER L/D

DO NOT RETRACT FLAPS

Ice Shedding 절차 반복



COLD Wx Operation 2/2

OAT (GND) / TAT (TAT) is 10°C (50°F) or below :

- visible moisture (clouds, fog with VIS 1SM (1600 m) or rain, snow, sleet, ice crystals...)
- ice, snow, slush and standing water is present on the ramps, taxiways, or runways.

PARKING

After engine shutdown and before shutting down electrical supply:

FLAPS / SLATS CONF FREE OF CONTAM

Perform a visual inspection to determine if the flaps/slats mechanism is free of contamination.

If necessary, perform decontamination.

Y ELEC PUMP pb AUTO F/O

G ELEC PUMP pb AUTO F/O

Y ELEC PUMP pb (guarded) ON F/O

G ELEC PUMP pb (guarded) ON F/O

SLATS / FLAPS RETRACT F/O

Monitor slats/flaps retraction on ECAM upper display

When slats and flaps are retracted:

Y ELEC PUMP pb (guarded) AUTO F/O

G ELEC PUMP pb (guarded) AUTO F/O

NORMAL PROCEDURE RESUME ALL



ENG ON Deicing in ICN

TOBT - 40min CTC KE ICN (사전신청, 결과확인)

ICN Deicing "Deicing Required ENG On Deicing"

ICN Apron "Req Pushback Deicing Zone xx **SQ2000**"

Pad Control Arrange Deicing Pad No. **REQ FULL BODY**

Ice Man Manage Deicing Process

Gate 시동후 (**FLAP UP, FLT CTL CHK** 없이)

PARKING BRAKE ----- **SET**

Report Parking Brake SET - > Ice Man

BROADBAND XMT sw ----- **OFF**

COMMUNICATION with GND ----- **ESTAB**

DE/ANTI-ICING FLUID TYPE ----- **CHECK**

CABIN PRESS MODE SEL ---- **CHK AUTO**

ENG 1 & 2 BLEED ----- **OFF**

Note : **AIR ENG 1+2 BLEED FAULT** – Disregard

APU BLEED ----- **OFF**

DITCHING pb ----- **ON**

Note : **CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV**

NOT OPEN, COND LAV + GAL VENT FALUT –Disregard

THRUST LEVERS ----- **CHK IDLE**

"A/C PREPARED FOR SPRAYING"

Report Ready for Deicing - > Ice Man

Cold Wx

START SPRAY **REQ DCL(CTC DEL)**

UPON COMPLETE SPRAY(**TIME CHK 1분**)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

PITOT STATICS (GND Crew) ----- **CHK**

GND EQUIPMENT ----- **REMOVE**

DITCHING pb ----- **OFF**

OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- **ON**

BROADBAND XMT sw ----- **ON**

5분후

APU BLEED ----- **AS RQRD**

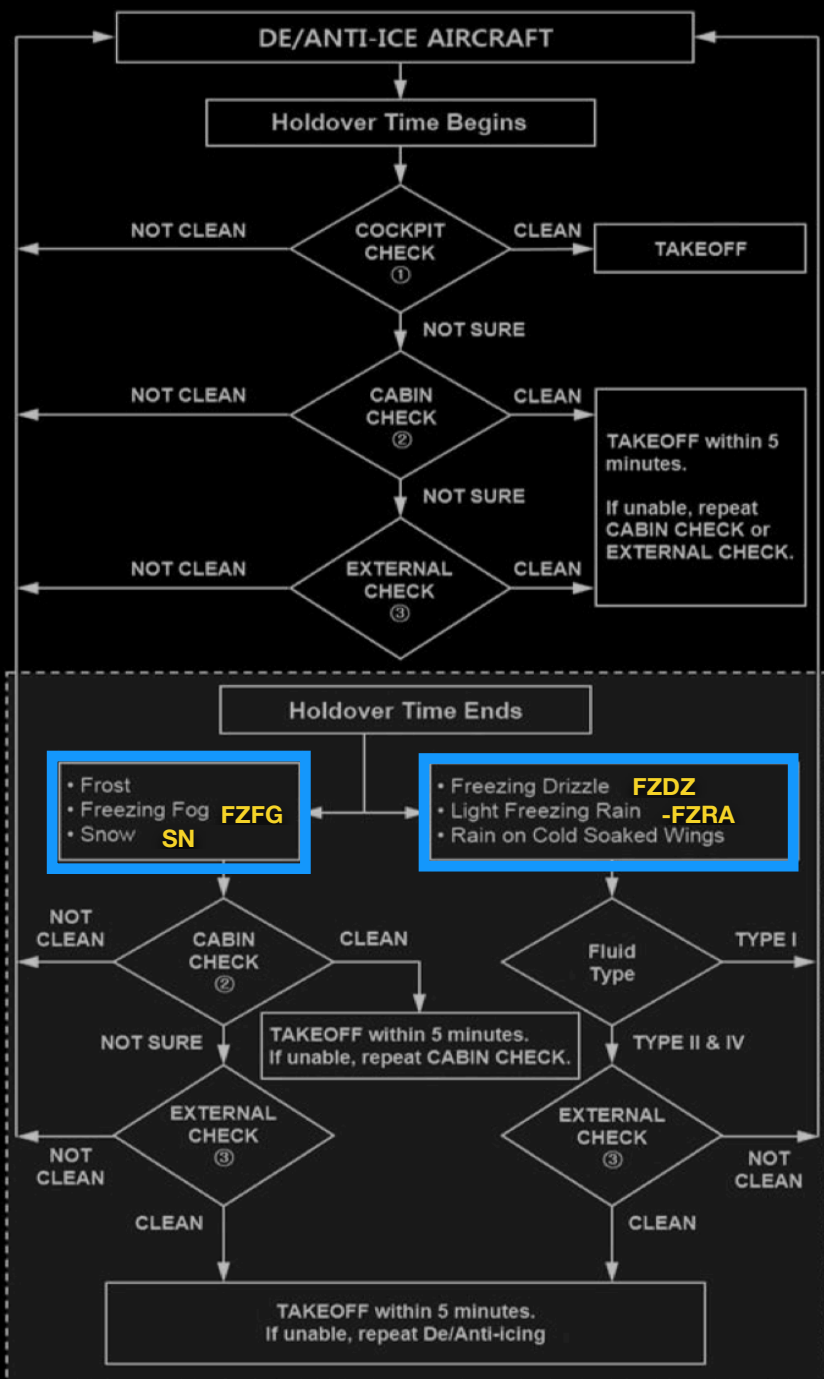
NORMAL PROCEDURE ----- **RESUME**

FLT CONTROL CHK, FLAPS UP



Decision Tree next page

TAKEOFF DECISION TREE



- **External Check = Second Deicing** (정비사가 보고 판단해야함)

- Heavy Ice Pellets(+PL), Heavy Snow(+SN), Hail(Small Hail 제외) (GR), Moderate 또는 Heavy Freezing Rain(FZRA, +FZRA) 이나 Heavy Freezing Drizzle(+FZDZ) 이 공항에 내리는 경우 이륙을 금지한다.



ENG OFF Deicing in GMP...

TOBT- 20min CTC KE GMP (PAD, New TOBT)

REQ DCL

Deicing "Deicing Required PADxxx" 5min TOBT

Apron "Req Pushback Deicing PADxxx"

Gate 시동후 (FLAP UP, FLT CTL CHK, APU OFF 없이)

Engine anti-ice sw ----- OFF

PARKING BRAKE ----- SET

BROADBAND XMT sw ----- OFF

COMMUNICATION with GND ----- ESTAB

DE/ANTI-ICING FLUID TYPE ----- CHECK

CABIN PRESS MODE SEL ---- CHK AUTO

(HL7524-7720) GND COOL ----- OFF

ENG 1 & 2 BLEED ----- OFF

APU BLEED ----- OFF

DITCHING pb ----- ON

Note : CAB PR FWD OFV NOT OPEN, CAB PR AFT OFV

NOT OPEN, COND LAV + GAL VENT FALUT -Disregard

ENG MASTER sw ----- OFF

SHUTDOWN CHECKLIST

"A/C PREPARED FOR SPRAYING"

UPON COMPLETE SPRAY(TIME CHK 1분)

용액과 마지막 용액 뿌린 시간 받고 적는다.

Holdover Time 결정!!!

Cold Wx

(HL7524-7720) GND COOL ----- OFF

PITOT STATICS (GND Crew) ----- CHK

GND EQUIPMENT ----- REMOVE

DITCHING pb ----- OFF

OUTFLOW V/V - CHK OPEN(ECAM press)

TIME CHECK 1분후

ENG 1 & 2 BLEED ----- ON

BROADBAND XMT sw ----- ON

5분후

APU BLEED ----- ON

NORMAL PROCEDURE ----- RESUME

Note : INIT B check & re-enter

PREFLT CHKlist -> Req STARTUP -> CHKlist

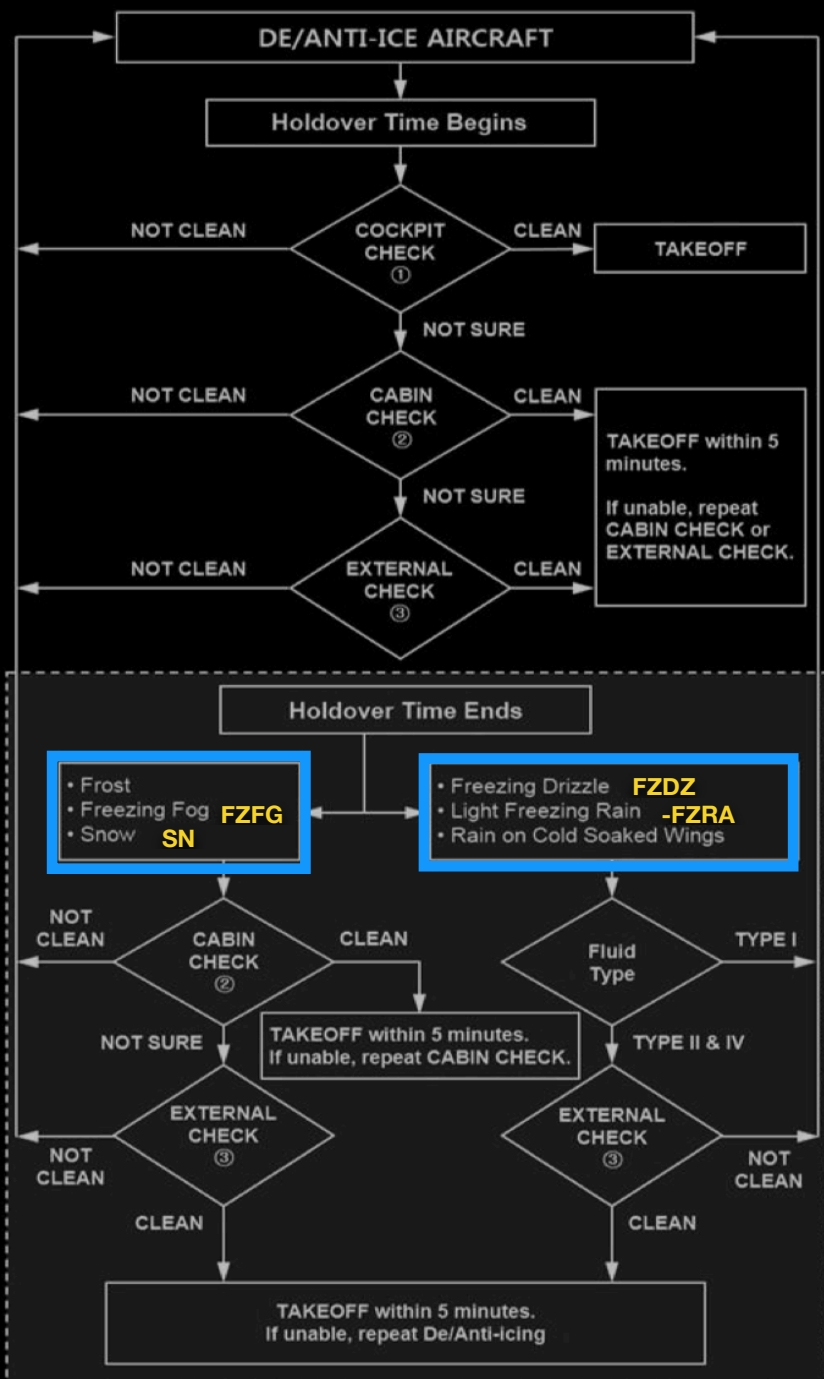
BOTH ENGINES ARE STARTED

FLT CTL CHK, FLAPS UP, APU as REQ



Decision Tree next page

TAKEOFF DECISION TREE



- **External Check = Second Deicing (정비사가 보고 판단해야함)**
 - Heavy Ice Pellets(+PL), Heavy Snow(+SN), Hail(Small Hail 제외) (GR), Moderate 또는 Heavy Freezing Rain(FZRA, +FZRA) 이나 Heavy Freezing Drizzle(+FZDZ) 이 공항에 내리는 경우 이륙을 금지한다.



QRH LIST Reference Only

AIR	BLEED 1+2 FAULT(APU BLEED ON) BLEED 1+2 FAULT(APU BLEED NOT AVAIL)
DOOR	COCKPIT DOOR FAULT
EIS	DISPLAY UNIT FAILURE EIS DISPLAY DISCREPANCY
ELEC	C/B TRIPPED ELEC EMER CONFIG SYS-REMAINING
ENG	ALL ENG FAIL ENGINE FUEL SYS CONTAMINATION ENG RELIGHT IN FLIGHT ENG STALL ENGINE TAILPIPE FIRE HIGH ENGINE VIBRATION ON GND - NON ENG SHUTDN AFT ENG OFF ONE ENGINE INOPER - CIRCLING APP ENG1(2) EGT OVERLIMIT
F/CTL	LANDING WITH SLATS OR FLAPS JAMMED NO FLAPS NO SLATS LANDING RUDDER JAM / RUDDER PEDAL JAM RUDDER TRIM RUNAWAY
FUEL	FUEL IMBALANCE FUEL LEAK FUEL LOSS REDUCTION FUEL OVERREAD GRAVITY FUEL FEEDING TRIM TANK FUEL UNUSABLE
L/G	LANDING WITH ABNORMAL L/G L/G GRAVITY EXTENSION
NAV	ABNORMAL V ALPHA PROT ADR 1+2+3 FAULT UNREL SPD INDICATION IR ALIGNMENT IN ATT MODE NAV FM / GPS POS DISAGREE ALL ADR OFF
SMOKE	SMOKE / FUMES / AVNCS SMOKE REMOVAL OF SMOKE / FUMES SMOKE / FIRE FROM LITHIUM BATTERY
WHEEL	WHEEL TIRE DAMAGE SUSPECTED
MISC	BOMB ON BOARD COCKPIT WINDSHIELD/WINDOW ARCING COCKPIT WINDSHIELD/WINDOW CRACKED DITCHING EMER EVAC EMER LANDING FORCED LANDING OVERWEIGHT LANDING SEVERE TURBULENCE TAILSTRIKE VOLCANIC ASH ENCOUNTER



SYSTEM RESET TABLE 1/2

CKPT CTL pb 3초

RESET BTN 1초

A-ICE	2 WHC	GND : A.ICE L(R) WSHLD(WINDOW) HEAT
AIR	ENG BLEED pb	AIR ENG 1(2) BLEED FAULT GND. : AIR ENG 1(2) BLEED NOT CLSD
	2 PACK CONT	PACK CONT RESET
COND	1 ZONE CONT	ZONE CONT RESET
APU	APU MASTER	Only APU SPD below 7%
AUTO FLT	REF FCOM	GND/Before TAXI : AUTO FLT A/THR OFF
	2 FCU	PART FCU, FCU
	2 FMGEC 2 FM	AUTO FLT FM 1(2) FAULT MAP NOT AVAIL
	MCDU OFF	MCDU RESET
BRKS	REF FCOM	GND : BRAKES RESIDUAL BRAKING WHEEL N/W STRG FAULT
CAB PR	2 CPC	FAULTY CPC RESET
COM	2 CIDS	Cabin Intercom Data Sys RESET
	2 ACP/AMU	ACP RESET
	RMP OFF	RMP RESET
	ELT RESET	ELT RESET
DATA LINK	1 ATSU	INVALID DATA or ATSU RESET
	1 AINS,1 CINS	AINS, CINS RESET
DOOR	2 PSCU	Proxi Switch Cont Unit RESET
EIS	DU OFF	ECAM, PFD, ND RESET
	3 DMC	DMC RESET
ELEC	EXT A,B OFF	GND : RESET EXT A,B
	3 BAT pb OFF	BATT RESET
	COMM, GALL pb OFF	GND : COMMER, GALL RESET
ENG	2 EIVMU	Eng Interface Vibration Monitor Unit RESET



SYSTEM RESET TABLE 1/2

CKPT CTL pb 3초

RESET BTN 1초

F/CTL	PRIM,SEC OFF	GND : PRIM, SEC RESET
	2 FCDC	FCDC RESET
	2 FLAP	GND : F/CTL FLAP SYS FAULT
	2 SLAT	GND : F/CTL SLAT SYS FAULT
FUEL	2 FCMC	GND : FUEL ZFW ZFCG DISAGREE
FWS	2 FWC, 2 SDAC	Waring Sys RESET
L/G	2 LGCIU	LGCIU RESET
MISC	VSC (Cabin aft C/ B)	RESET VACUUM TOILET SYS (Blue pb)
COND	1 ZONE CONT	ZONE CONT RESET
SMOKE	2 SDCU	Smoke Dectect RESET
VENT	2 VENT CONT	RESET VENT CONT
	1 AEVC	RESET VENT COMPUTER

